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The Global Cashless Payment Revolution: a look at Africa and Europe

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First and foremost, thank you God for keeping my family, friends, and loved ones safe during this pandemic. Big ups to my support system for keeping me on track during this process.

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Abstracts

Préface

L'objectif de ce mémoire de licence est d'explorer les possibilités d'un avenir sans argent liquide, et si notre société est prête pour une telle transition. J'examinerai comment un tel système serait mis en œuvre, les éventuels revers économiques et sociaux qui pourraient être rencontrés en cours de route, ainsi que les avantages que le passage au sans numéraire pourrait apporter à la société. Selon plusieurs professionnels des domaines de la Fintech et de l'économie, la fin de l'argent liquide comme moyen de règlement des transactions sera inévitable ; la question est simplement de savoir quand et surtout comment.

Je vais revenir sur l'histoire de la création du système d'échange de biens et de services, pour pouvoir ensuite analyser les caractéristiques de notre système monétaire actuel régi par la monnaie fiduciaire et ses implications économiques. Plus loin, je détaillerai le cas de deux pays très différents, l'un sur le point de devenir la première société sans numéraire, et l'autre une société où l'argent liquide est le moyen de paiement le moins fiable. L'objectif principal de cette comparaison est d'analyser les données disponibles sur la base d'articles scientifiques, d'articles de presse, de statistiques et d'enquêtes afin de paver les bases d'un avenir sans monnaie physique.

En outre, il y aura une discussion sur la viabilité de la technologie blockchain pour soutenir la transition vers une société sans numéraire. J'évaluerai les avantages et les inconvénients, ainsi que les implications macroéconomiques et sociétales de l'utilisation de monnaies numériques s’aidant de la technologie blockchain dans le but de compléter les espèces physiques, et éventuellement de les remplacer.

Resumen

El objetivo de esta tesis de licenciatura es explorar las posibilidades de un futuro sin dinero en efectivo, y si nuestra sociedad estaría preparada para tal transición. Se examinará la forma como se aplicaría ese sistema, los posibles reveses económicos y sociales que podrían encontrarse en el camino y los beneficios que una transición sin dinero en efectivo podría aportar a la sociedad. Según varios profesionales de los campos de la tecnología y la economía, el fin del dinero en efectivo como medio de liquidación de las transacciones será inevitable; la pregunta es simplemente cuándo y, sobre todo, cómo.

Se repasará la historia de la creación del sistema de intercambio de bienes y servicios, para luego analizar las características de nuestro actual sistema monetario regido por el dinero fiduciario y sus implicaciones económicas. Más adelante se detallará el caso de dos países muy diferentes, uno a punto de convertirse en la primera sociedad sin dinero en efectivo, y el otro una sociedad en la que el dinero en efectivo es el medio de pago menos fiable. El principal objetivo de esta comparación es analizar los datos disponibles sobre la base de artículos científicos, artículos de prensa, estadísticas y encuestas para sentar las bases de un futuro sin dinero físico.

Además, se debatirá la viabilidad de la tecnología de cadenas de bloques para apoyar la transición a una sociedad sin dinero. Se evaluará las ventajas y desventajas, así como las implicaciones macroeconómicas y sociales del uso de monedas digitales utilizando la tecnología de cadenas de bloques para complementar, y posiblemente reemplazar, la moneda física.

Abstrakt

Das Ziel dieser Bachelorarbeit ist es, die Möglichkeiten einer bargeldlosen Zukunft zu erforschen, und ob unsere Gesellschaft für einen solchen Übergang bereit ist. Ich werde mich mit der Umsetzung eines solchen Systems, den möglichen wirtschaftlichen und sozialen Rückschlägen, die auf diesem Weg auftreten können, sowie mit den Vorteilen befassen, die ein bargeldloser Übergang für die Gesellschaft bringen könnte. Nach Ansicht mehrerer Fachleute aus den Bereichen Fintech, Finanzen und Wirtschaft ist das Ende des physischen Bargelds als Mittel zur Abwicklung von Transaktionen unvermeidlich; es ist einfach eine Frage des Wann und Wie.

Dazu werde ich auf die Geschichte unserer Systeme zum Austausch von Gütern und Dienstleistungen zurückgehen, um die Merkmale unseres gegenwärtigen, vom Fiat-Geld regierten Währungssystems zu erklären. Im weiteren Verlauf werde ich den Fall von zwei sehr unterschiedlichen Ländern erläutern, von denen das eine auf dem Weg ist, die erste bargeldlose Gesellschaft zu werden, und das andere eine Gesellschaft, in der Bargeld die am wenigsten vertrauenswürdige Zahlungsform ist. Das Hauptziel dieses Vergleichs besteht darin, die vorliegenden Daten auf der Grundlage von wissenschaftlichen Abhandlungen, Zeitungsartikeln, Statistiken und Umfragen zu analysieren, um eine Grundlage für die Auswirkungen einer bargeldlosen Zukunft zu schaffen.

Darüber hinaus wird es eine Diskussion über die Durchführbarkeit der Blockkettentechnologie zur Unterstützung des Übergangs zu einer bargeldlosen Gesellschaft geben. Ich werde die Vor- und Nachteile digitaler Währungen auf der Grundlage der Blockkettentechnologie bewerten, mit dem Ziel, physisches Bargeld zu ergänzen und möglicherweise irgendwann zu ersetzen, sowie die makroökonomischen und gesellschaftlichen Auswirkungen der Nutzung dieser Technologie

Abstract

This Bachelor thesis aims to explore the possibilities of a cashless future, and if our society is ready for such a transition. This paper will look at how such a system has been implemented; the possible economic and social setbacks encountered along the way, in addition to the benefits going cashless could bring to society. According to certain observers in the fields of Fintech, finance and economics (Manur, 2016), the end of physical cash as a means of settling transactions is inevitable; it is merely a question of when and how.

This paper will examine the history of our systems of exchange of goods & services to explain the characteristics of our current monetary system governed by fiat money. Further along, this paper will explore the case of two very different countries, one on the path of becoming the first cashless society, and the other where cash is the least trusted form of payment but the only available one. The main goal with this comparison is to analyse the data at hand based on scholarly papers, news articles, statistics and surveys to lay down a basis of the implications of a cashless future.

In addition, there will be a discussion on the viability of the blockchain technology to support the transition to a cashless society. This paper will evaluate the advantages and disadvantages of digital currencies powered by blockchain with the objective to compliment physical cash, and possibly eventually replace it, as well as the macroeconomic & societal implications of using this technology.

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List of abbreviations

ACH – Automatic Clearing Houses

AD – Anno Domini

ATM – Automatic Teller Machines

BC – Before Christ

BOA – Bank of America

BTC – Bitcoin

CAGR – Compound Annual Growth Rate

CIC – Cash in circulation

COVID-19 – Coronavirus Disease of 2019

EUR – Euro

GDP – Gross Domestic Product

ICA – Interbank Card Association

INR – Indian Rupee

MDC – Movement for Democratic Change

MSE – Micro and Small Enterprises

MSME – Micro, Small and Medium Enterprises

NFC – Near Field Communication

POS – Point of sales

PoW – Proof of Work

P2P – Peer to peer

SEK – Swedish Krona

TZS – Tanzanian Shilling

UK – United Kingdom

USA – United States of America

USD – United States Dollar

ZANU-PF – Zimbabwe African National Union-Patriotic Front

ZWL – Zimbabwean Dollar

Introduction

Since the beginning of time, cash has existed and been at the centre of how everything works: to feed, grow, colonise, fund wars, bribe and much more; it is hard to imagine a world without it. However, recent trends have suggested that people seem to be moving towards *cashless* alternatives to pay for goods and services and live their lives. A quick search of the words 'cashless payments' on any search engine displays clickbait headlines intended to incite curiosity and wonder.

Figure 1 - Article published by the Washington Post 2020

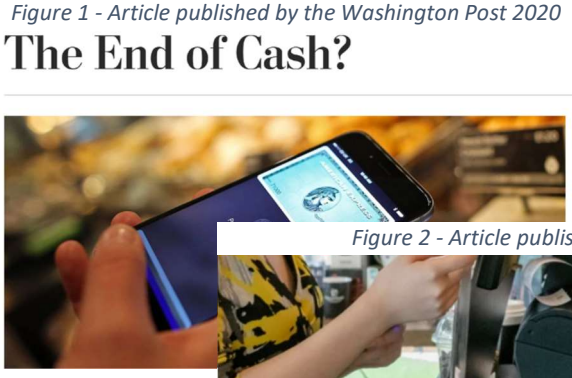
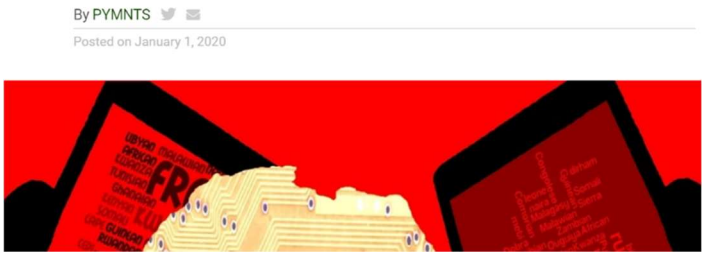


Figure 2 - Article published by Financial Times 2018



Figure 3 - Article published by PYMNTS 2020



What are these cashless alternatives, and why are they becoming so popular? Let us have a quick look at where they came from.

The first payment card was introduced in the 1950s in the USA for the founding of the Diner's Club in New York, allowing the cardholders to use them in individual retail shops. They worked on the system of credit, where the card acted as an "instant lender" the user had to pay

back at the end of each month. The ATM quickly followed, allowing the flexibility and convenience of getting cash anywhere at any time.

A couple of decades later, debit cards came to revolutionise the game in the 1980s in the USA and quickly spread throughout the world. They were different from earlier cards because they were equipped with magnetic chips which allowed the sum to be instantly debited from the total of funds in the user's bank account & directly transferred to the payee. The popularity of debit cards initiated the implementation of a worldwide debiting system built on the credit card system.

From there, the system grew exponentially in popularity: since 2010, card transactions have started to overtake cash transactions in certain parts of Europe (Boesenach, 2018). Transactions are processed at lightning speed, allowing millions of customers worldwide to enjoy spending and consuming in our globalised capitalistic society of today.

So, the relevant question to ask is why? Why part with cash? The answer can be as simple as the natural forces of the market of demand and supply: people are using less cash than before. The complicated answer: it depends on many factors.

Issuers of currency, usually Central Banks, may notice the decreased demand for physical currency, and when demand falls for an item, it becomes more expensive to provide it; thus, their incentive is to produce less of it to avoid incurring losses. People using less cash over time is part of the evolution of the industry of payments, just as the labour industry had evolved in the 18th century with the automation of production lines; sure, it created jobs, but it also reduced them in the farming industry for example.

Cash handling/printing services and cash-essential industries in certain countries are at the front of the 'fight' against cashless because it is their bread and butter.

The elderly and those with cognitive/physical disabilities are with these industries: they are not ready to let physical cash go because either

they do not want to 'evolve' their payment method or are simply unable to. The physical characteristic of cash assures them that they are holding a failproof form of payment in case banks fail and any of their savings stored digitally vanish. Physical money allows them to be ordinary active citizens, especially if they are unable to open a bank account or access digital alternatives, which is a reality for many.

However, large parts of the young population are either indifferent or in agreement with the transition to cashless transactions. Having a light, easy and portable method of payment either in the form of a card or through their mobile appeals to them, especially those closely following trending techs.

The remaining group are worried about privacy. If cash were to disappear and all transactions to go through a bank account, it would leave a valuable data trail behind with every purchase made, exploitable by the bank for commercial and unethical purposes.

There is cause for concern in the moving away from physical money. The concentration of transactions on one method of payment is not recommended because what if the system supporting that method becomes momentarily unavailable? Millions of transactions would be cancelled, and the economy would be forced to press 'pause' for a moment, ensuring substantial economic consequences. The speculative theory became real-life when VISA[1] services went down for 10 hours across Europe in June 2018. In a statement issued by VISA to the UK parliament, they calculated that "during the period of the service disruption, 51.2 million Visa transactions were initiated and sent to Visa 's European systems for processing. Of these, 5.2 million (10%) failed to process correctly" (Hogg, 2018). The disruption left people unable to go about their day to day, socialise and pay merchants, and also left small businesses unable to receive payments. Looking at the numbers, the disruption may not seem immense, but it points to a bigger problem regarding large centralised payment

[1] A card payment processor company

networks. When users realise they are unable to pay with their card, they lose trust in its ability to transact and will go to lengths to hold physical cash, which is what happened in the UK during the outage where several ATM's ran out of money (Lewis & Kozhani, 2020). Even so, keeping a diverse portfolio of methods of transacting may not be beneficial in the long run either, especially if the demand for one significantly exceeds the others.

To understand the debate surrounding the cashless revolution, we need to look back in history. This paper will follow this structure: the invention of money starting with its deep roots in the past and the evolution it went through, the rising 'cashless revolution' dubbed 'The War on Cash' by some, and the economics governed by physical money. These three parts will be beneficial to understand the profile of two countries chosen to demonstrate going cashless in two manners: the case of Sweden and Zimbabwe. A survey will aid in analysing trends regarding the opinion of citizens residing in different parts of the world about going cashless to further add to the cases of these two countries.

Furthermore, this paper will discuss the current policies in place to accommodate these disruptive forms of payment relating to both country cases and the literature review. There will be an expansion on the rise in digital currencies, the technology of blockchain, and its feasibility to aid the transition towards cashless societies.

Nonetheless, the future is not as far as we think, and some countries may already be well on their way to eliminating centuries of tradition.

Literature review

The History of Money

What is money, and where does it come from?

Money is globally understood and known as "something generally accepted as a medium of exchange, a measure of value, or a means of payment: such as officially coined or stamped metal currency …, money of account and paper money"[2]. For the context of this thesis, the shortened definition is as follows: a current medium of exchange and payment in the form of minted coins and paper[3].

One important notion to remember is that money in itself is nothing: it is the value that people place on it that matters. "Money derives its value by being a medium of exchange, a unit of measurement […] for wealth[4]". The more people trust money, the more valuable it becomes.

To earn a better understanding of the future of money, we need to look at the past, where its' invention took place before the beginning of written history. The first findings regarding our predecessor's knowledge on the concept of transacting were found in ancient scriptures of Mesopotamia, where the idea of someone 'owing one' was recorded in the form of debt, at a time where coins did not exist yet. Anthropologist David Graeber proposes in his book '*Debt: The First 5,000 years*' that this thinking of 'owing one' led to the idea of 'owing one unit' of debt and further on developed the barter system (Tett, 2020).

To barter is "to trade by exchanging one commodity for another**:** to trade goods or services in exchange for other goods or services[5]", where animal skins could have been exchanged for papyrus papers, who knows. The barter system lasted approximately 2000 years until a physical representation of one unit of value in the form of cowrie

[2] Merriam-Webster dictionary 2020
[3] A Mint is a facility where coins and paper money used as currency are manufactured
[4] Investopedia's "The History of Money: From Barter to Banknotes" 2019
[5] Merriam-Webster dictionary 2020

shells and beads in Africa was recorded. This use of representative money, defined as "any medium of exchange […] that represents something of value but has little value of its own" can also be linked to the invention of coins in the ancient empires of China, Babylon, Egypt and India (Gawra, 1996). Scriptures from the Chinese empire in 1000 BC suggested they were the first to transition to an early form of the monetary system, where they started "…using miniature replicas of … tools cast in bronze[6]" overtime changing the shape of these replicas to a more circular form, recognisable as a coin. Empires in Europe followed suit using similar shapes made of valuable commodities like gold or silver to transact.

Subsequently, China issued the first currency notes during the Tang Dynasty[7] , which ended up being used almost exclusively to buy goods and services around the world. Europe transitioned to paper money in the 16th century, where banks issued banknotes that resembled in function to today's cheques. They could be taken to the bank to be exchanged for their face value in coins, or to pay for goods directly. The only difference from today's paper money was that it was issued by banks and private institutions, as opposed to the government's central bank today.

The shift to paper money increased the amount of international trade because banks and the ruling classes started buying currencies from other nations, which prompted the creation of the first currency market. The stability of a country affected the value of its currency and the ability for them to trade on the international market; currency wars often occurred to affect the value of the competitor's currency, their purchasing power on the global market and the price of their goods.

Fast forward a couple of centuries to the invention of the first credit card in 1949, when sir McNamara took clients out to eat in a restaurant

[6] Investopedia's "The History of Money: From Barter to Banknotes" 2019
[7] 618 AD – 907 AD

in New York City and realised he had forgotten his wallet at home. He figured there must be another way to pay for entertainment or travels than always requiring to have enough cash on hand. The next year, he created a small cardboard charge card, the first Diners Club card and used it to pay at that same restaurant. "It was the first time in modern history a customer purchased what he could afford, not what he had enough cash on hand for"(Cochrach, 2019). He later convinced 27 local restaurants to accept the card as payment, and a year later, 42 000 merchants were using the platform.

In 1957, American Express (an express mail business at the time) saw the potential in the charge card business and launched the first Amex cards to be used with the company's traveller checks. The issue now was that people had too many different cards for too many different things. Besides, most of the accounts attached to the cards were charge accounts, meaning they were expected to be paid off at the end of each month which restricted a large part of the population unable to make a pay-check last this long. A year later, Bank of America saw an opportunity in the problem and issued test cards with a 300$ credit limit ready to use. Keep in mind back then there was no system of verification like today to own a card or to protect it with a pin or password, which led to a significant amount of frauds; eventually, financial controls were put in place.

Mid 1960s, BOA was thriving: it was signing agreements with other banks to issue cards that could be used on the same network in the USA and around the world. Wells Fargo followed in BOA's steps and launched the interbank card under the Interbank Card Association (ICA), later renamed Mastercard in 1979. BOA saw the benefit in separating the card service from the mother company, which led them to do so in 1976 and named the independent entity VISA. The debit card shortly followed, gaining in popularity over the credit card, and soon spread throughout the whole world. Both Mastercard and Visa evolved to becoming top card payment system processors and signed agreements with banks over the globe to provide the payment cards.

In 2018, VISA card transactions accounted for 44.84% of total card transactions worldwide; Mastercard generated 24.46% of total transactions worldwide (Details & Transactions, 2020).

At the turn of the Millennium, digital technological developments led to the birth of cryptocurrencies with Bitcoin in 2009. By definition, cryptocurrencies are "any form of currency that only exists digitally, that usually has no central issuing or regulating authority but instead uses a decentralised system to record transactions and manage the issuance of new units, and that relies on cryptography to prevent counterfeiting and fraudulent transactions[8]".

In the beginning, Bitcoin's value was so low, holding 1 BTC was like having a unit worth zero. For the next three years, its value slowly climbed as economies recovered from the 2008 global financial crisis, and people shifted away from banks to explore new forms of payment. Its value started skyrocketing in 2014 reaching a top value of 1,000$ per 1 BTC before crashing to levels close to 200$; its value later increased and stabilised over the years, before reaching an ultimate high in December 2017 of 17,000$ per 1 BTC. Today (28th April 2020), 1 BTC is worth 7,764$ and can be used to purchase high-value goods or services around the world.

In technical terms, Bitcoin's system rests on the combination of cryptography and blockchain. Cryptography is a method of protecting information through the use of codes. A blockchain is a continuously expandable list of data records called blocks which are linked together using cryptography. Blockchains differ in three aspects: control, permission and consensus; and face the following trilemma: decentralisation, scalability and security, which means no blockchain can 100% be all three. Bitcoin operates on a completely decentralised system where the blockchain is public, permission-less and functions on the PoW[9] consensus algorithm. This means anybody can

[8] Merriam Webster dictionary
[9] Proof of Work

participate in updating the blocks without requiring any permission from any authority. Miners are rewarded when they prove to have mined the block.

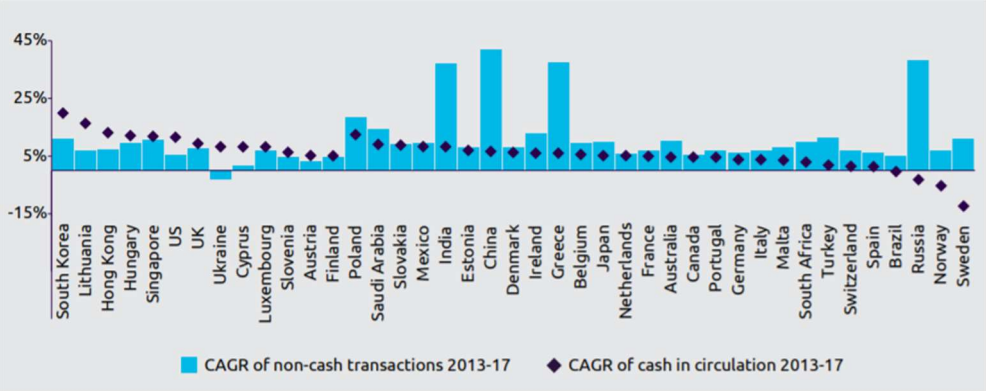
The rise of Bitcoin in 2015 prompted the creation of other cryptocurrencies, both public and private, advertising key appealing characteristics: tamper-proof and decentralisation. Tamper-proof because once a block is mined, it is time-stamped and impossible to modify; decentralisation because there is a reduced risk of a single point of failure and no central authority to govern the network. However, cryptocurrencies remain very volatile as they are exclusively driven by demand and supply is limited due to certain restrictions in the code.

Having gone through the history of money from its beginnings preceding the barter system to today's emergence of digital currencies, cash still remains the most established and widely used payment instrument in the world. However, disruptive technology in payment methods is a growing phenomenon estimated to rival with cash in the future. Some are dubbing it 'The War on Cash', and this is the next part.

The War on Cash

As of January 2020, Euro banknotes & coins in circulation in the Euro Zone were estimated at over €1.3 trillion[10], with an annual increase of 6.7% from January 2008 to December 2017. The value in circulation

Figure 4 - Growth of cash in circulation vs non-cash transaction, by country, 2013-2017



[10] ECB Statistical Data Warehouse

slowed to 4% in 2018, but the Euro remains the highest in the world. Across the globe, cash in circulation has been consistently growing from 4-7% annually over the last five years, shown in figure 4 (Capgemini, 2019).

So why the mention of a war on cash? What is it, and why is it relevant to this topic?

In technical terms, the war on cash encompasses the set of policies deployed by government agencies to suppress the use of paper currency. The main objective of these policies is to shift transactions made in cash to payment cards and bank services, which leave a data trail accessible when required by law enforcement and tax authorities. Secondary objectives include the fulfilment of monetary policy goals of central banks to push nominal interest rates to zero, which necessitates the increase in the cost of cash storage, incentivising people to spend.

One of the most commonly known 'tactics' of this war is the obligation to declare any cash amount on your person above a specified value when crossing national borders. Other 'tactics' include the abolition of big bills, the placement of a legal ceiling on cash transactions (up to 5000€ in Europe), and the requirement of banks to report to authorities when someone withdraws large sums near or over the indicated suspicious amount.

Harvard Economist Kenneth Rogoff considers the expression' war on cash' an exaggeration, specifically in regards to the word cash which he deems unqualified. He is an advocate of the war 'only on big bills' and does not agree to pursue an entirely cashless society. "…criminals use large notes, so anyone who uses them might be a criminal." seems like a compelling argument in favour of banning high denomination notes (White, 2018). Doing so could indeed make life harder for criminal enterprises; however, it could also make it harder for everyone else because not only criminals use big bills.

The war on big bills with the intent of reducing criminal activity is arguable across the spectrum of currencies and their denominated notes. The abolition of the €1000 bill can be debated because of how much its value represents in illegal markets. Trying to ban a 10,000TZS note is illogical mainly because, its value represents almost nothing on illegal markets and its abolition would significantly inconvenience people. It would be like trying to ban €50 notes; people would not be pleased. To further explain why, let us take a look at India.

In November 2016, the government announced a demonetisation policy rendering the 500 and 1000 INR notes invalid, giving people 50 days to exchange them for the newly minted 500 and 2000 INR notes. The two previous notes alone made up 86% of the currency circulation and 62% of the money stock in India. The unexpected policy was intended to inflict losses on tax evaders & criminals, but instead created consequential collateral damage on honest users of the currency. The initial announcement caused panic, ensuing shortages of the currency which led to endless hours of people queuing. Paired with the inefficiency of the government to supply currency in time, it eventually deeply impacted the cash-dependent unbanked half of the Indian population. Shortages of banknotes meant labourers did not get paid, farmers could not sell their produce and people died not being able to afford medical care.

Advocating for the War on Cash in countries where most of the population does not have a bank account and heavily relies on cash is not very logical. All it does is direct the war on all the people who use cash, not just the ones who use it with criminal intentions. Coercive anti-cash policies decrease freedom and reduce the welfare of peaceful individuals who like to use cash. These policies compromise financial privacy and harm the livelihood of small & micro-businesses.

However, a nation's desire to reduce criminal activity is not actually the main driver on the war on cash. Several of the following factors have contributed to the phenomenon over the past couple of decades.

Drivers of the War on Cash

In the 1970s, banks in the USA became overwhelmed with the massive volume of paper checks they were receiving. This led to the creation of ACHs[11], "…a network that coordinates electronic payments and automated money transfers…"[12] which allowed money to move between banks without using paper checks. This included the direct deposit of wages, payments from businesses to suppliers & vendors and automatic payment of recurring bills such as insurance or electricity.

In the 1990s, e-commerce was new and on the rise. Websites like AuctionWeb (now eBay) were receiving envelopes of cash because payments online were unavailable. The arrival of PayPal at the end of the decade revolutionised the online shopping experience, it "..quickly became the platform of choice for those wanting to make online purchases, or … just to send money to friends or family" (Cochrach, 2019). The inefficiency of using cash for online transactions led to the rise of new types of electronic payment methods in the next two decades, notably mobile phones and then smartphones, allowing two disruptive forms of currency: mobile payments and digital currencies.

Mobile payments are transactions for goods or services made through portable electronic devices such as cell phones, smartphones or tablets. Examples include ApplePay developed by Apple Inc., GooglePay by Google LLC and M-Pesa developed by Vodacom Group Limited.

The case of M-Pesa is interesting to look at because, unlike GooglePay and ApplePay, it did not require the user to have a bank

[11] Automated Clearing House
[12] The Balance 2020

account. Launched in 2007 in Kenya, it kickstarted the mobile money revolution in sub-Saharan Africa. It significantly contributed to the financial inclusion of populations unable to open a bank account but with easy access to mobile phone. The service allows them to 'deposit' and 'store' money on their account registered under their phone number, just like you would add airtime, and use that to pay for transactions by sending the money to another user's account, a vendor for example. M-Pesa provided an easy, accessible alternative to cash to populations unable to access formal banking services. One of the first countries where M-Pesa was exported to was Tanzania via Vodacom in 2008, where it was met with success. At the time, in a population of 42 million, there were only 13 million registered mobile cellular subscriptions; after the introduction of M-Pesa, the number of registered mobile cellular subscriptions grew 13.8%[13] on average per year, reaching over 40 million by 2018 (Mbiti et al., 2016). With a 70% penetration of mobile phones in Tanzania in 2018, M-Pesa had 8.2 million active users, 42% of the mobile payment market and over 1.2 trillion transactions per month. Competition in the mobile payment market has led to significant innovation; Vodacom Tanzania partnered with Mastercard in 2018 to provide the M-Pesa virtual card linked to the user's account allowing them to pay for online transactions whether local or international. With more than 12.4 million[14] customers today, that number is expected to grow and offer a complimentary service to cash for Tanzanians in the future.

Lastly, digital currencies are types of currencies available on a distributed database on the internet and allow for instantaneous and borderless transfers of ownership (Al-Laham et al., 2009). Examples include virtual currencies, cryptocurrencies and Central Bank issued digital currencies.

The 'famous' cryptocurrency Bitcoin was created and first mined in January 2009. In the beginning, it held little real purchasing powers,

[13] My own calculations – see appendix
[14] Wikipedia “Vodacom Tanzania” – see bibliography for more info

but one of its main characteristics, anonymity, made it find a niche in the underworld. Ross Ulbricht opened an online black market called Silk Road which became a drug trafficking haven where any substance could be purchased with Bitcoin. When he was arrested two years later, more than 1.2 million transactions with a value of 9.5 million BTC had already occurred on the platform, which set the pattern and the reputation for cryptocurrencies. Cash does provide anonymity, but anonymity on a global scale thanks to the internet held much more value, especially in black markets.

Now, Bitcoin can be used to buy many different goods, from luxury items to real estate. Other digital currencies have emerged on the decentralised ledger powered by the blockchain technology, offering extra privacy like Monero or for the use of big corporations like Libra for Facebook. The ones most likely to succeed and endure in such a competitive environment need to solve specific real-world problems and appeal to the public and the proper target market.

Why the war on cash matters?

The Global Findex database reports data on financial indexes (how adults borrow, save etc…); in their 2017 report, they found that 94% of adults in high-income economies have a bank account, compared to 63% in low- or middle-income economies. Having access to financial services is an important step towards reducing both poverty and inequality. Innovations in Fintech, especially with the arrival of mobile payment services, show great promises to bridge the gap in financial inclusion, encourage commerce and make transactions safer for users.

The War on Cash, for the most part, seems to be waging in high-income economies, where the majority of the adult population has access to financial services, diversified payment methods and seems to have less interest in cash. However, it is essential to ask who this war on cash benefits; is it in the public's interest or the interest of tax authorities?

It can be argued that it is instead a war on privacy as the prime objective of the policies aforementioned is for all transactions to leave an audit trail trackable by tax authorities. Besides, some debate that the war could benefit central banks anticipating to use more negative interest rates in their monetary policies to boost the economy out of a recession. This could affect people who have deposited most of their wealth in a bank account and see it being charged negative interest, thus incurring losses. The existence of cash allows them to retrieve their money in physical notes and wait until interest rates are positive again, which is counterproductive to a central bank's policies. At most, the war on cash seems to have a distant relative- the war on drugs. If taken to the extreme, civil liberties could be overlooked in the criminalisation, surveillance and prosecution of private activities.

The war on cash, as exaggerated as some consider the phrase, is a trend that is unlikely to die down. Looking at high-income economies in Europe, numbers speak for themselves; consumers demand less and less cash finding digital alternatives more convenient and more secure than their physical counterpart. Technology has been the main driver of this war, significantly contributing to lowering the price of essential financial services and democratising the access to banking in low- and middle-income economies, thus allowing more people to participate in the economy and lift themselves out of poverty.

Having now looked at influencing factors of this war, it is essential to understand how embedded cash is in our economy.

The economics of cash

The transition from the barter system around 3000 BC to the monetary one in 1000 BC improved the efficiency, scalability and speed at which people could buy and sell goods and services. As a reminder, a monetary system is a set of policies and institutions by which governments create money in an economy. This system and the money used in it evolved over centuries allowing the differentiation between 3 types: commodity money, commodity-based money and fiat money.

Commodity money is the earliest form of currency in a monetary system, where commodities or precious metals such as gold or silver were used to pay for transactions. The introduction of representative money swiftly followed because people found it safer and easier to carry a certificate/token which represented the commodity, that was in the bank. These certificates were issued by the bank and could be exchanged for goods and services. Overtime people grew to trust the representative money and use it more frequently and substantially. Governments eventually decided that money no longer needed to be backed by a commodity as it had been for centuries because it would always be limited by that commodity's finite supply on Earth thus potentially and eventually halting a nation's growth. This led to the creation and widespread use of fiat money.

Fiat money is a government-issued currency that is not backed by a physical commodity but rather by the government that issues it[15]. Fiat money does not hold the intrinsic value that representative or commodity money have. Its' value comes from its existing and growing money supply and being declared as a legal tender[16]. The national currency is the legal tender in the majority of countries and has value because the government says so, and people trust it and use it to settle

[15] Partial citation Investopedia “Fiat Money”

[16] the means to settle public and private debt and to settle financial obligations

transactions. Today, most of the fiat money can be found in the form of bank balances and records of card purchases.

Money vs currency

There is a common misconception that money and currency, which are used interchangeably, mean the same thing. The two are close relatives, but differences become apparent when we talk about their economic implications.

In its early forms, money was tangible and often took the form or was backed by valuable commodities/metals like gold and silver. They are considered valuable because they are scarce; they have a finite stock on Earth. Today, money has three definite functions: it is an intangible medium of exchange, a store of value and a unit of account which takes an entirely numerical form and holds purchasing power over a long period.

The arrival of fiat money created a system where notes and coins minted by a central authority became referred to as currency. Currency is not a store or a measure of value, it is tangible in nature and acts as a medium of exchange through the minted notes and coins; fiat money is a currency that holds no intrinsic value, unlike money. Authorities have bestowed value upon it by declaring it legal tender and expecting people to trust it and use it. If all trust on fiat money were to disappear, its' value would plummet to zero, and anything stored as such would be lost, such as savings in a bank account for example. In an ideal system, fiat money needs to be durable, portable and divisible; uniformity, acceptability, limited supply and hard to counterfeit would be a bonus for the perfect society. Limited supply is essential because scarcity creates value, which is needed to compliment the one given by authorities.

Economic indicators and monetary policies

To discuss the path of the cashless societies, it is important to include how a monetary system would function without physical currencies; a

system where the current fiat money would only be found in the form of bank balances and card payments. We need to understand the characteristics and functions of today's monetary system and why individuals or central banks are not ready to part with currencies just yet.

Central banks control how much currency is in circulation in a country, to make sure its value stays stable over time and contributes to any of their macroeconomic goals. Regulating supply goes back to the basics of market forces in economics. When the supply exceeds the demand, its' 'price' (value) drops and vice versa. Money supply is an economic indicator and a teller of a country's fiscal policies. When a central bank controls the money supply, it inherently controls the purchasing power of the people because scarcity creates value. When the value of a currency is said to be decreasing, it results in the erosion of purchasing power, formally known as inflation.

In economics, inflation is described as the sustained increased of general prices in an economy over a period of time. Inflation serves as an economic tool put in place by issuers of currency to encourage people to spend their money instead of saving, with the ideology that money today will be worth less tomorrow. A healthy level of inflation usually fluctuates around 3-5%, anything above is labelled as high inflation. When this occurs, the people's purchasing power decreases because wages don't adjust as fast as the prices of goods and services due to certain rigidities.

Anything below 0% is called deflation, defined as the general decrease in the price level of goods and services in an economy. It is not good for an economy because people hold on to their money with the expectation that prices will be lower tomorrow. This ends up slowing down economic growth in a country because spending drops, production drops and unemployment subsequentially increases, digging the economy deeper in a crisis.

Then there is the case of hyperinflation, defined as the "rapid, excessive, and out of control general price increase in an economy"[17] where wages aren't able to keep up with this price increase, therefore significantly impacting the people's purchasing power. One of the causes of hyperinflation is related to an excessive rise in the money supply.

The quantity theory of money states that inflation will increase when money supply growth exceeds the expansion of real economic activity, assuming the velocity of money[18] remains the same. Expectations play a significant role in perpetuating higher prices during bouts of hyperinflation, and the effect of those expectations on money and inflation is amplified relative to other influences such as the business cycle. To reverse hyperinflation is to restore the people's belief in the government's ability to implement fundamental monetary and fiscal policy changes to ensure a stable disinflation (Federal Reserve Bank of Dallas, 2011).

These three types of inflation can be a result of a multitude of events such as fluctuations in exchange rates, demand-pull or cost-push factors, national debt and money supply. When inflation is mainly caused by the latter, monetary policies are usually put in place to contain the sustained increase in prices. A monetary policy "…refers to the actions undertaken by a nation's central bank to control the money supply to achieve macroeconomic goals that promote sustainable economic growth."[19]. Macroeconomic goals include low levels of unemployment, stable inflation and GDP.

Central banks tend to use interest rates to govern these policies. If they are contractionary, CBs will increase interest rates and decrease the money supply precisely to control and reduce inflation; if they are expansionary, CBs will lower interest rates and increase the money supply thus creating inflation, to boost aggregate demand. A liquidity

[17] Partial citation Investopedia “Hyperinflation”
[18] The number of times it changes hands
[19] Partial citation Investopedia “Monetary Policy”

trap can occur if people don't spend or invest even when interest rates are low, rendering the expansionary monetary policy useless and further digging the economy in a crisis.[20]

The role of money has weaved itself into complex relations with pillar components of our economy and the way everything functions today. The evolution of our monetary system now governed by fiat money seems deeply embedded and necessary for our lives to go on.

[20] Partial citation Investopedia “Liquidity trap”

Country profiles

The Case of Sweden

The Kingdom of Sweden, a Scandinavian country situated in Northern Europe, has made headlines over the recent years for its path to possibly becoming the first cashless society of the 21st century.

Since 2007, Sveriges Riksbank[21] recorded a steady decrease in the use of cash in Sweden. The value of the CIC of the Swedish Krona (SEK) had dropped 50% in 10 years as innovations were made, robberies happened, and time went by. In 2017, 18% POS transactions in Sweden were made in cash (Arvidsson et al., 2018), the lowest figure in Europe compared to its neighbours Finland with 33% and Germany with 55% (Esselink & Hernández, 2017).

In his book 'The Swedish Route to the Future of Cash Payments', author Niklas Arvidsson claims "The decrease in the use of cash happens because it is legally possible and because merchants and consumers tend to prefer other solutions". To fully understand Sweden's path to becoming perhaps the first cashless society in the world, it is important to look a little further back in history.

History

Swedish history can be traced back as far as 8000 BC with early human settlements, the Vikings in 800 AD and the Kalmar Union of the 13th century. The timeline relevant to this topic is the minting of the first coin in 1604, the riksdaler, to further promote trade which was booming in the region. The creation of the first bank, Stockholm Banco, in 1656 helped to develop the system of borrowing and lending. Stockholm Banco launched credit notes to expand their crediting network and allow for individuals to invest, businesses to borrow and to fund industrial operations throughout the Kingdom. These credit notes were backed by securities such as valuable commodities (gold, silver) However, the bank started printing more notes than securities

[21] The Swedish Central Bank

available to back them, which created a bubble that eventually burst, effectively closing the bank in 1664. The bursting of the bubble shone a light on how institutions had to be wary of the quantity theory of money.

The Kingdom then decided to set up the Swedish Central Bank in 1668 and to learn from the first bank's mistakes. They pegged the national currency to gold, the US Dollars and British pounds to facilitate trade and instil confidence in the currency.

Moving a couple of centuries forward, the economic scene in Sweden post World War II was blooming. Businesses were growing, wages were increasing, people were spending, and the quality of life was progressing. At that time, nearly all the transactions between individuals were made in cash. The rise of new technologies led the banks in Sweden on a path of innovation to increase the number of people they were serving. They introduced the first transaction bank account, revolutionising modern banking. Employers would deposit salaries in employees' accounts at less cost than if they were to give it to them in cash directly. Employees benefited from being able to withdraw their wages freely and other services that the banks offered. The popularisation and globalisation of credit/debit cards arrived in the 1980s-90s in Sweden, complementing the beginning of e-commerce platforms. However, a significant part of the population was still heavily using cash for a lot of their transactions, and banks, together with card companies, wanted to change that.

Influencing factors to reduce cash

The 2000s was a turning point on the views of cash in Sweden; the cash industry was struck by unanticipated events. Robberies of banks, retailers and cash deposits grew in frequency and popularity early in the decade, causing substantial losses to the industry and sparking campaigns to reduce the use and stockpiling of cash in public transport, banks, retailers, etc. The robbery of 2009 where SEK 39 million (around EUR 4 million) were stolen in a way worthy of any 21st-

century action movie reignited the debate on the safety of cash and a plan to rethink its use, even if people weren't ready to let it go just yet. To quote Arvidsson, "cash is a strong artefact and institution in our society", national currencies further validate the sovereignty of nations, and their physical form is regarded as a symbol, a representant of the nation and a failproof manner of obtaining goods and services.

Around the same time, Sweden introduced new tax incentives to stimulate the economy, to reduce tax evasion and the amount of business that operated on black money. Black money is defined as "income illegally obtained or not declared for tax purposes[22]". Informal jobs in the building and private sector had been paid for in black money for decades. The introduction of these incentives influenced private persons to generate invoices or pay with any means other than cash. Merchants were also affected by these incentives as it obligated them to declare all sales at cash registers, thus making it more difficult for them to avoid paying taxes. This encouraged them to turn to other payment forms, such as card, as it made the accounting process much simpler and less costly too.

In 2012 came the turning point in the debate of cash. Fanaxia, a cash transit company, filed for bankruptcy and sequentially caused merchants all over the country to lose a lot of money. Cash transit companies offer services that, as the name indicates, handle, store and transport cash depots between bank branches, large retailers and premises holding large amounts of cash. After Fanaxia declared bankruptcy, the market of cash-handling companies was left with this power vacuum, lowering competitivity, thus allowing players to increase their fees. This event further intensified the concerns and uncertainty of merchants towards cash and incited them to seek other methods of receiving payments. With the arrival of Swish later that year, citizens now had a variety of choices to pay for transactions.

[22] Lexico definition “Black money”

Swish is a P2P[23] mobile payment service in cooperation with national commercial banks in Sweden, allowing the transfer of money in real-time between the user's bank and the recipient's bank. With the original intent of targeting trendy young individuals, it soon moved to markets, sports club and other places where cards or cash were impractical. It facilitated the exchange and paved the way for instant transactions. Today, Swish has more than 7 million active private users, and 230 000 companies accepting it as a means of payment.

Lastly, reports on macroeconomic and socioeconomic indicators were factors to influence the debate on cash. According to 'The cost of Consumer Payments in Sweden', a paper published in 2012 by the central bank of Sweden, the socioeconomic cost of cash is higher than that of plastic money[24]. Although the study is a bit outdated as it is based on 2009 findings, it is the only one that offers substantial insight into the cost of payment methods of different acting parties of society.

Socioeconomic costs can range from payment intermediary costs such as those for insurance or technical equipment to expenses related to waiting time for individual customers at a POS.

"There are indications that society's cost per payment in Sweden has fallen since then. For example, the payment solution Swish […] has emerged to take over from cash payments" (Sverigerisksbank, 2019). The cost per payment using a particular payment system depends on how many payments are made due to high fixed costs in that process. As the number of payments increases through the process, let us say debit cards, each payment is cheaper, thereby reducing an individual's cost per payment using a debit card more significantly than any other payment system. This effect further proves that the price of cash increased, as mentioned when Fanaxia filed for bankruptcy. There is room for improvement even though the cost per payment of Swedes

[23] Peer-to-peer
[24] Debit, credit, charge cards

has reduced, such as more innovation and competition in streamlining the efficiency of instant payments.

To summarise, the main factors explaining the reduction of cash in Sweden are

- → The market dynamics of the supply and demand for cash
- → Innovation in card payments especially with the contactless features
- → Innovation in mobile payments with Switch and NFC contactless service
- → The increase in e-commerce

Transition

Even though Sweden is a developed well-functioning economy in Europe, it is important to look at socio-geographic characteristics of the nation. According to 2019 statistics, Sweden has a relatively small population of just over 10 million which rests on an area larger than 400 000 km², geographically more important than Germany, Japan and Iraq to name a few. But if you compare the population density, which gives an idea on how 'close' people live, Sweden's population density per square meter is 25.4 (Sweden Statistics, 2020), compared to 233 for Germany (Federal Statistical Office Germany, 2020). This means that people in Sweden live geographically far away from each other, which makes it challenging to supply cash handling services at adequate prices. This further increases the risk of a digital divide, not only in rural areas but also in larger cities majority populated by groups that are not ready or unable to follow the transition.

The elderly with physical/cognitive disabilities, immigrants, the physically/mentally disabled are the major constituents of these groups that already have trouble to access basic cash services. Accompanied by cash-essential and cash-handling industries, they are the primary advocates asking to keep cash or slow down its decrease in Sweden.

Conclusion

"A payment service must realise network effects and interoperability to become valuable for both payers and payees. This depends on the number of users […] in the system" Arvidsson mentions when speaking about the innovation of payment services (Arvidsson, 2019). The decrease in the interoperability of cash has led to a tipping point that may possibly be attained in the next decade or so because the network value of cash is too low in relation to the costs for merchants to continue accepting it, or for users to continue using it.

"Today we see a strong substitution effect where mobile payment services—like Swish and iZettle—replace cash payments…This has led Sweden to become the country in the world with the lowest use of cash." (Arvidsson, 2019). The partnership between iZettle and Situation Sthlm in 2013, a street newspaper sold by the homeless, in a project to test cashless methods of receiving payments further sparked the debate on the transition of cashless.

Sweden is inevitably moving towards possibly becoming the first cashless society in the next decade. The challenge for policymakers is to make sure that few people as possible are left behind during this transition. That is, provide adequate solutions for those unable to join the digital movements and adapt these technologies for the groups of people at risk of being separated by the digital divide.

The Case of Zimbabwe

The Republic of Zimbabwe has a complicated history of money, liquidity and opting for cashless solutions, very different from the Kingdom of Sweden.

Geography and history

Landlocked in the sub-Saharan region of Africa between Mozambique, Zambia, Botswana and South Africa, Zimbabwe has a total land area of over 39 million acres, 85% of which is used for agricultural purposes. According to the latest data, the agricultural sector generates approximately 12% of Zimbabwe's GDP and employs 67% of the total labour force[25]. The principal cash crops are tobacco with one of the highest exports in Africa, and cotton mainly contributing to the flourishing textile industry. Under British colonial rule since the 1880s, Europeans settled and controlled most of the arable land for decades. When Rhodesia gained independence from the United Kingdom after 100 years of colonial rule in 1980, a government under Prime Minister Mugabe came in power. It treaded cautiously through the policies put in place by the previous government. The economy started to decline in 1990 due to several factors, accelerating significantly in the early 2000s with Mugabe's land reform program.

The agricultural sector was dominated by the minority Zimbabweans of European descent who stayed after independence; they owned approximately 1/3 of all arable land and contributed to 40% of the country's exports. In an effort to 'honour' the promises of the 1979 Lancaster House Agreement[26], The government bought and (mostly) seized by force a lot of these lands to be redistributed to war veterans and landless farmers. As a result, at least 30% of the workforce became unemployed, and individuals with political connections ended up claiming these lands. However, having no experience in farming,

[25] World Bank data 2018

[26] Partial citation Wikipedia (see bibiliography) - the Lancaster House Agreement was an effort to more equitably distribute land between black subsistence farmers and white Zimbabweans of European ancestry, who had traditionally enjoyed superior political and economic status. The programme's stated targets were intended to alter the ethnic balance of land ownership.

they were unable to maintain productivity. They thus contributed to the decline of the sector, food shortages ensued, and total exports dropped significantly (*Zimbabwe: A country study*, 1983).

Politics and economic reforms

After the land reform of 2002, Mugabe's government was becoming increasingly unpopular and aggressive towards critics and opposition leaders. Freedom of the press was restricted, and several newspapers were shut down. In the midst, the International Monetary Fund decided to suspend all loans because of- mainly the country's inability to repay previous loans, the highly criticised land reform of 2002 and human & political rights infringement.

Zimbabwe was already in a bad place: economic mismanagement and corruption had dug the country deep in a crisis, leading to record levels of unemployment and inflation. With no official numbers, institutions estimated that 4/5 of the total workforce was unemployed. Zimbabweans moved away to neighbouring countries to find work. The official government estimated inflation reached 8000% in late 2007, 100,000% in early 2008 before peaking to 10 million % in the summer. Without any access to loans or bailouts, the Zimbabwean currency, the Zimbabwean Dollar (ZWL) suffered and became practically and literally worthless.

The 2008 elections were filled with violence and truly brought Zimbabwe to its lowest. Mugabe's party (ZANU-PF[27]) had lost parliamentary majority for the first time and polls indicated they were losing the election against the opposition party MDC[28]. A run-off election was held later that year because neither parties had collected over 50% majority; MDC supporters and candidates were harassed and beaten by ZANU-PF loyalists and supporters, causing the presidential candidate of MDC to withdraw from the election. Mugabe won but international observers deem the election was neither free nor

[27] Zimbabwe African National Union-Patriotic Front – political party founded in 1987 and ruling since the independence of Zimbabwe

[28] Movement for Democratic Change – political party founded in 1999 in Zimbabwe

fair. Since MDC had won the majority of seats in parliament, a power-sharing agreement was signed, and an inclusive government formed. Amidst the political turmoil, the living standards of Zimbabweans plummeted, shortages of food occurred, the public sector quickly deteriorated and led to the deadly cholera epidemic of 2008.

To halt inflation and provide some economic stability, the new government suspended the Zimbabwean dollar and implemented a multicurrency system in 2009. Restructuring programs were implemented in the agricultural and mining sectors to improve Zimbabwe's exports and contribute to economic growth. After the 2013 elections filled with suspicions of fraud, ZANU-PF won again and petitioned to change the constitution in a way that would keep Mugabe in power until 2023. The economy suffered in 2016 with droughts leading to weak exports, thus a shortage of cash as the people over-relied on one currency (the USD); bond notes were then introduced to complement the USD and curb this overreliance. However, the discontent of the people as regards to slow economic growth, soaring unemployment, and the government trying to bring back the ZWL sparked demonstrations against ZANU-PF and Mugabe that same year.

In 2017, a 'coup' was orchestrated by the military to pressure Mugabe to resign, which led to Mnangagwa to take office as interim president, before being officially elected in office in 2018.

To this day, people still debate the root causes of Zimbabwe's fall. From the point of view of ZANU-PF supporters: it was unfair economic sanctions, the failure of the British government to honour the 1979 Lancaster Agreement regarding the transfer of land ownership to black Zimbabweans, and a western plot to get Mugabe out of power that drove Zimbabwe to mayhem. From the point of view of everyone else, especially critics of the government: it was the land-seizure program, economic mismanagement and widespread corruption that even other African leaders condoned.

Going cashless: unpopular but potentially beneficial

Having looked at Zimbabwe's history, its downfall and slow recovery, we can better understand the benefits and challenges going cashless has brought to the nation and what the future reserves.

Zimbabwe's banking sector developed significantly with the arrival of foreign banks and technological advancements of the 21st century. However, the emergence of cashless alternatives made life harder instead of easier. For those who have access to financial services, long wait lines and high transactions fees to withdraw or pay are the norm (Nyoni, 2018). Those residing outside of urban areas have more difficulty in accessing banking facilities.

Although there are many advantages linked to cashless alternatives, there is also an unwillingness to change. It seems to stem from a lack of trust in the banking sector, a lack of understanding of the concept of plastic money[29], exploitive charges in the use of plastic money and poor network services. The lack of confidence in the banking sector is primarily due to savings going up in smoke during the hyperinflationary period of 2008 (Nyoni et al., 2017). Trust and confidence are critical factors for the success of any monetary policy because currency in itself has no value except the one bestowed upon it by the people's belief in its utility.

According to an empirical analysis conducted in 2018 in Zimbabwe by Professor Nyoni, a survey with 567 responses gave her the following insights:

- → 83% of the respondents believe that a cashless economy is beneficial because it facilitates faster transactions
- → 77% of the respondents believe that a cashless economy could reduce cash-related affairs of corruption

Respondents also agreed that going cashless could reduce inflation as it exercises control over inflationary pressures such as the

[29] Debit, credit, charge etc… cards

fluctuations in the money supply. However, 96% of respondents said that cashless payment facilities were not available everywhere, especially in rural areas, which shows how underdeveloped the banking system still is. 54% of the respondents also agreed that the current payment facilities accessible complicate the way of doing business because of high transactions costs and long queues at the bank. Other answers to the question as to why they aren't ready to fully embrace cashless economies were current high transaction costs and a general lack of financial knowledge and skill (Nyoni, 2018).

A focus group conducted by Sakarombe and Marabada in 2017 in the city of Harare shed a light on how much the people don't trust the government. Many have the feeling that the government's policy inciting the use of plastic money serves only their interest to use that money as they see fit. There is also the fear that hyperinflation could return especially with the introduction of the bond note in 2016, which has resulted in people hoarding foreign currencies in cash.

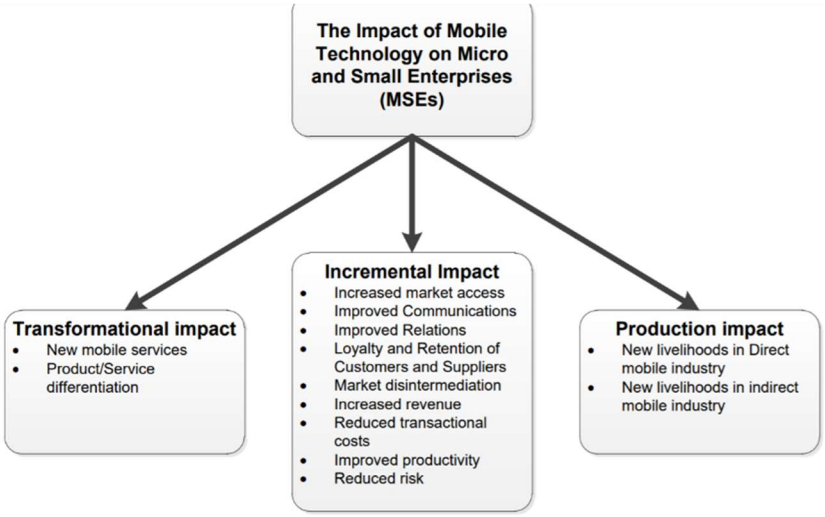
"The research by Patel and Amin (2012) alluded the viewpoint that plastic money is the roadway to a cashless society. In their discussion, plastic money has become a common way of transacting, and it has become the best way of controlling money laundering and ensuring tax legislation." (Nyoni et al., 2017)

However, plastic money isn't the only cashless method of payment; mobile payments figure in the list too. A mobile payment system allows for payment for goods or services through a portable electronic device such as a tablet or a cell phone. Commonly known examples include ApplePay, Paypal, M-Pesa, Airtel Money and EcoCash, just to mention a few.

"Mobile technology has helped in shaping our new global society. In some of emerging economies, more people have access to mobile technology while they lack necessities such as clean water, electricity, sanitation and bank accounts (Kelly & Minges, 2012)" (Perekwa et al., 2016).

It is increasingly becoming recognised as contributing to the development of social, economic, political and environmental settings in developing countries. The introduction of M-Pesa in East Africa in the 2000s is an excellent example of it. As well as being more accessible than traditional banking facilities, mobile technology also significantly contributed to the development of MSMEs[30].

Figure 5 - Impact mobile technology on MSEs (Perekwa et al., 2016)



Referring to figure 5, mobile technology has had three types of impacts on MSMEs: transformational, incremental and productional. Concerning Zimbabwe, the MSME sector dominates the economy as it provides 5.7 million jobs and generates an annual turnover of 7.4 billion USD (World Bank, 2012). It significantly contributes to alleviating poverty due to increased market access, creation of employment because of improved productivity, thus developing the economy.

However, it is essential to ask how cashless alternatives could help empower the Zimbabweans and foster economic growth, as the current system is creating more problems than it is solving. Also, cashless options could help combat inflation which stands at 676%

[30] Micro, Small and Medium Enterprises

today, a 500% increase since the beginning of the year due to the COVID-19 pandemic and other geographical factors[31].

Conclusion

Zimbabwe has been a cash-based economy for a long time, and most of the business which pays people is informal and unbanked. The main hurdles restricting people to access and use these cashless alternatives include:

- → High service charges, unavailability of cashless platforms everywhere and a general lack of skill/knowledge on these alternatives, thus financially excluding a big part of the population
- → High costs of access as POS machines not profitable for businesses
- → Elevated lack of trust in the government and the banking system due to past economic mismanagement

The government should first work to regain the trust of the citizens before trying to incentivise or coerce them into any policies. A decade of economic mismanagement, visible corruption and no foreseeable intent of change on the government's part will make it hard for them to praises about the benefits of a cashless economy.

Secondly, to ensure a properly functional cashless economy, there needs to be adequate infrastructural facilities in place, such as sufficient POS machines, ATMs and telephone towers. The current banking system is clearly inefficient and needs to be further developed to accommodate users better. Transaction fees of commercial banks are high, potentially because of the uncertainty of hyperinflation returning hovering around in the air, which is why stability is essential. If people always expect for the system to fail, they will make sure to make the most of it before it does.

[31] Reserve Bank of Zimbabwe data for March 2020

Zimbabwe has a liquidity problem as hyperinflation in the past has been linked to the Reserve Bank increasing the money supply to pay off debt and fund government spending. The multicurrency system did bring stability, but the people's overreliance on the USD caused a shortage of cash. However, there seems to be hope in mobile payment technologies. The introduction of M-Pesa in East Africa helped give access to an informal type of banking to millions of people, allowing them to set up their business, control their finances and contribute to the economy. For Zimbabwe, mobile banking and payment technologies could help control inflationary pressures influenced by the money supply and restore some faith and stability in the monetary system.

Research methodology

The following section describes the approach and research method of this paper, and links theories with my findings of the relevant academic topics. This thesis aims to achieve a comprehensive understanding of monetary systems of today and if cashless solutions will play a lead role in their future.

The research of this thesis is based on empirical evidence collected from news articles, academic papers and financial reports published between 1983 and 2020 from multiple sources both found online and offline. The topics addressed range from Global History, Economics and Blockchain technology, including the idea of cashless societies potentially becoming our future. The key takeaways from some of these academic works mentioned in the literature review are centred on the socioeconomic hardships and benefits going cashless could bring to certain countries, with an emphasis on the deeply rooted aspects of cash in their societies.

In addition to reviewing the literature, a survey created on Google Forms was conducted and distributed online through private social media accounts (WhatsApp, Instagram & Twitter) to extract both quantitative and qualitative data. It is structured in the following order:

- → An introduction explaining the purpose of the survey, including:
  - Specific instructions to ensure accurate and pertinent answers
  - The time duration of completion.
- → The first part of the survey aimed to collect demographic information on the participants, the frequency in the use of specific payment methods and their opinion on said methods.
- → The second part evaluated the basic knowledge of the participants about digital currencies and if they owned or traded any, and their opinion on their potential to possibly replace cash in the future.

→ A "thank you" message to end the survey, indicating the responses have been recorded.

The survey consists of single choice answers, multiple-choice answers and free answers, which contribute to a qualitative data result. There were no specific conditions or additional materials required to complete the survey successfully. The choice to conduct it online was influenced by the need to obtain quick raw data.

The quantitative answers provide mostly demographical statistics, such as age and country of residence, and the frequency with which they use a particular payment method. The qualitative answers provide insight to evaluate trends of payment preferences, and the opinion of participants on payment solutions of the future according to their country of residence.

The main target group of this survey were people of generation Z and Y from Europe and Africa regardless of their occupation or gender. The age was put forward because certain of the works analysed in the literature review emphasised the differences in mindsets of different ages about payment preferences. Also, it could be beneficial to discuss the possibility that generation Z will be the one to significantly influence the future of payment methods in the coming decades. The participants in the survey are a random sample group predominantly made up of young people between the ages of 16-25 years[32] from Europe and Africa. Unfortunately, participants between the ages of 26-40 years[33] were a minority which may cause some limitations and inaccuracies to the conclusion.

The findings have an inductive character, meaning a hypothesis was generated based on the answers of the participants. This hypothesis will provide an understanding to the general trend of payment

[32] Generation Z
[33] Generation Y – also known as “Millennials”

preferences and opinion to be compared to the general conclusion of the literature review. The approach cannot be statistically analysed due to certain limitations in the size of the sample and the nature of some of the questions asked; nevertheless, it remains systematic and random, thus appropriate.

The result of this empirical research aim to prove that the younger generations in Europe are more inclined to diversify their payment options subsequentially using less cash. That same generation residing throughout the African continent are aware of the shift towards cashless solutions and use them, but still prefer to use cash due to certain socioeconomic factors.

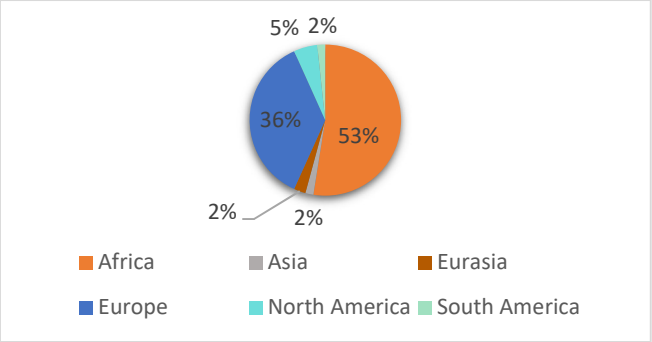
Research analysis

A survey distributed through Social Media platforms with a target audience mostly made up of international students collected 132 responses from around the world. The goal of the survey is to provide fundamental insight into people's payment preferences and their knowledge of digital payment alternatives to physical money. The survey asked both direct and indirect questions about the cashless revolution in terms of their country of residence. The answers will be analysed separately into two groups based on the age of the respondents.

Respondents aged 16-25 (generation Z)

118 responses from all around the world were recorded in this age range. For clarity, I decided to separate the answers depending on the different continents mentioned in figure 7. Due to the significantly smaller size sample of respondents from Eurasia, Asia, North & South America, answers will be concentrated on respondents from Europe and Africa.

Figure 6 - Provenance of participants aged 16-25 years



The survey did not inquire about the gender, the occupation or social status of the participants; only their age and country of residence are relevant for the demographics. 100% of European participants have a bank account as opposed to 82% of African participants. Majority of both know what online payment systems are and have used at least one.

Looking at figure 7, we can see that 34% of respondents who reside in Africa deem cash as essential in comparison, opinions vary from essential to non-essential for European residents. We can further see this variation by looking at the frequency of use of certain payment methods for certain activities.

Figure 7 - Perceived usefulness of cash (16-25yrs)

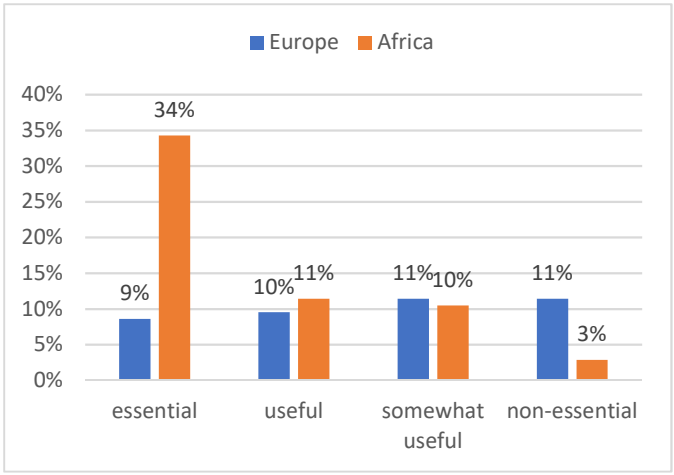
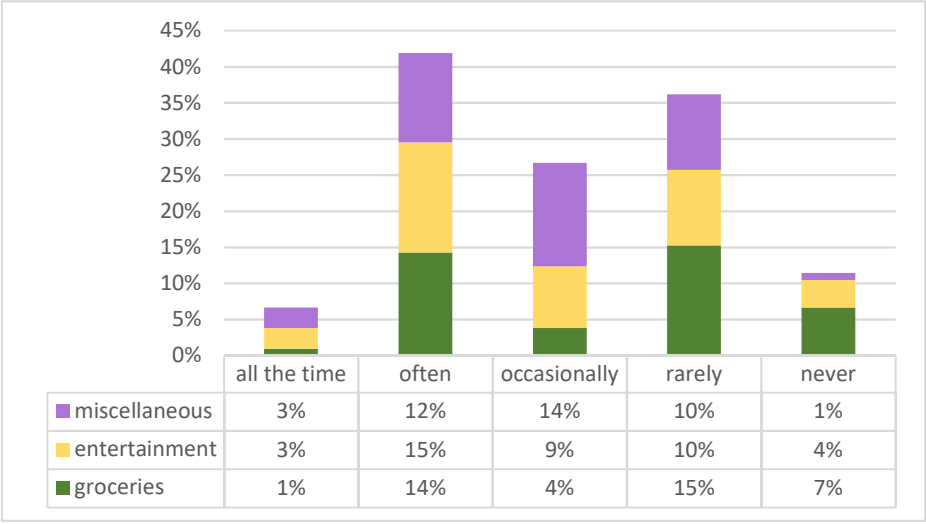


Figure 8 shows that European respondents tend to rarely use cash for groceries, to use cash for miscellaneous expenses[34] and often for entertainment occasionally[35].

Figure 8 - Cash use Europe (16-25yrs)



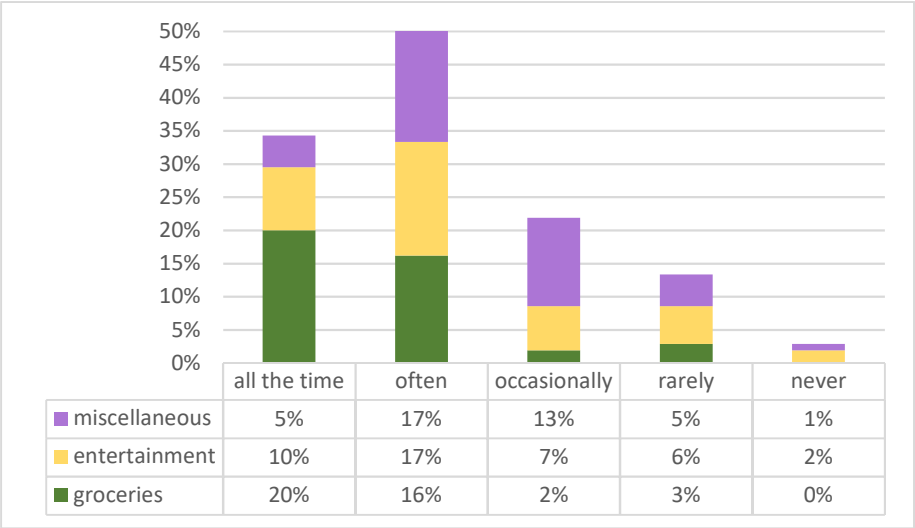
	all the time	often	occasionally	rarely	never
miscellaneous	3%	12%	14%	10%	1%
entertainment	3%	15%	9%	10%	4%
groceries	1%	14%	4%	15%	7%

[34] Miscellaneous expenses can include (but aren't limited to), any purchases that aren't on your grocery list and which don't figure in the entertainment list mentioned above. Examples include goods like tobacco or services like cleaning.

[35] cinema, bar, club, restaurant etc...

If we take a look at figure 9, which shows how often they use card for these three categories, we can observe a more apparent preference. The majority use card for groceries all the time, use it often for entertainment, and between often and occasionally for miscellaneous expenses.

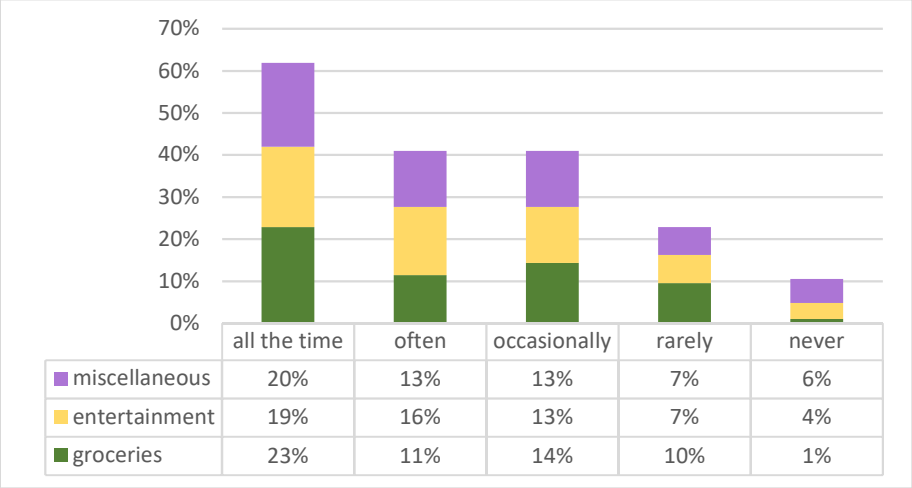
Figure 9 - Card use Europe (16-25yrs)



	all the time	often	occasionally	rarely	never
miscellaneous	5%	17%	13%	5%	1%
entertainment	10%	17%	7%	6%	2%
groceries	20%	16%	2%	3%	0%

The results for African respondents shown in both figures below are as expected from their opinion on cash seen in figure 7. The majority seem to use cash all the time for all three categories, with few using it occasionally. Even though 82% of African respondents have a bank account and so a payment card, looking at figure 10, they still use it

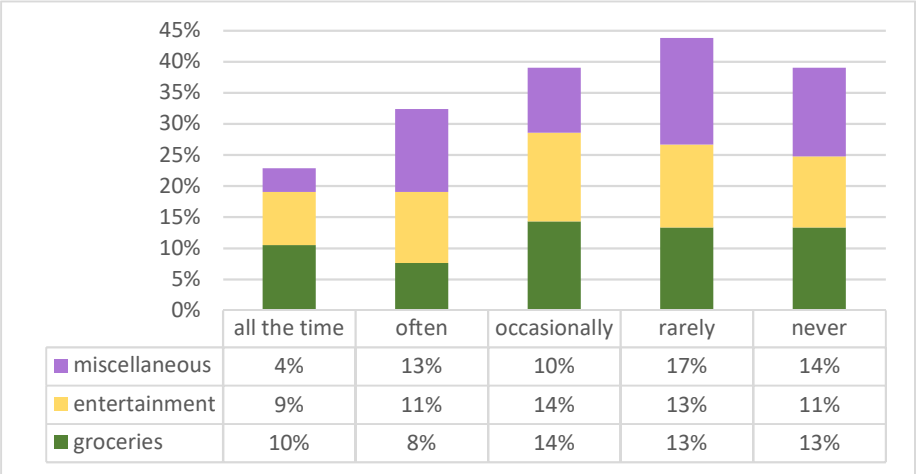
Figure 10 - Cash use Africa (16-25yrs)



	all the time	often	occasionally	rarely	never
miscellaneous	20%	13%	13%	7%	6%
entertainment	19%	16%	13%	7%	4%
groceries	23%	11%	14%	10%	1%

occasionally or rarely for all three categories. This could be due to the limited availability of card readers in POS.

Figure 11 - Card use Africa (16-25yrs)



	all the time	often	occasionally	rarely	never
miscellaneous	4%	13%	10%	17%	14%
entertainment	9%	11%	14%	13%	11%
groceries	10%	8%	14%	13%	13%

Regarding the participants' knowledge of digital currencies, we can see that majority know what they are, and mostly European respondents trade or own digital currencies as opposed to the majority of African respondents who do not.

Figure 12 - Knowledge about digital currencies (16-25yrs)

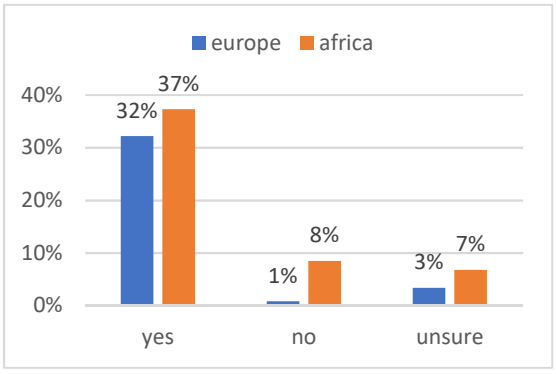
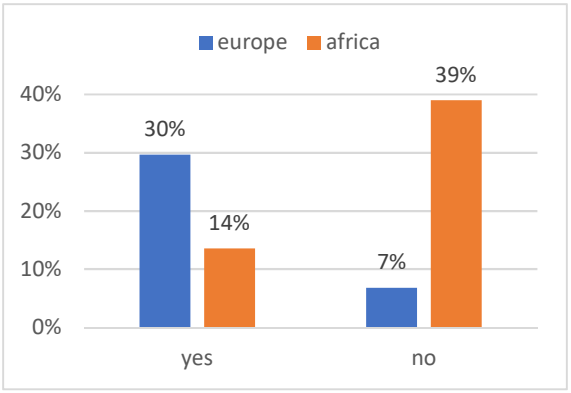
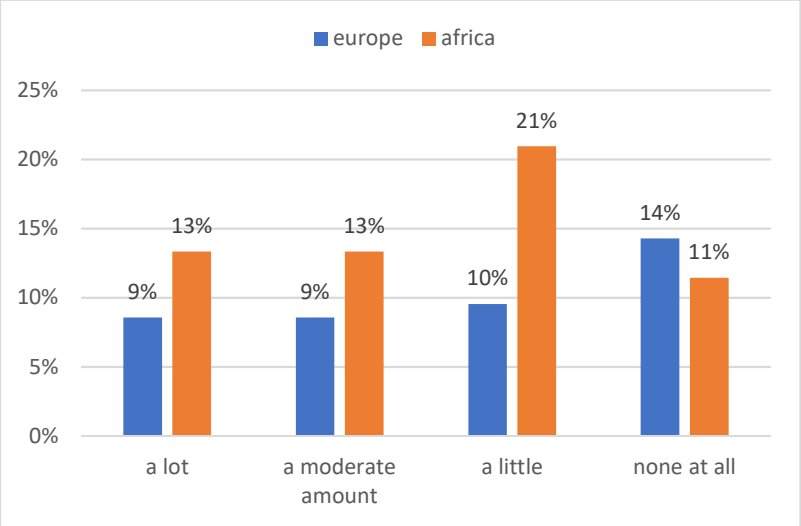


Figure 13 - Trade or own digital currencies (16-25yrs)



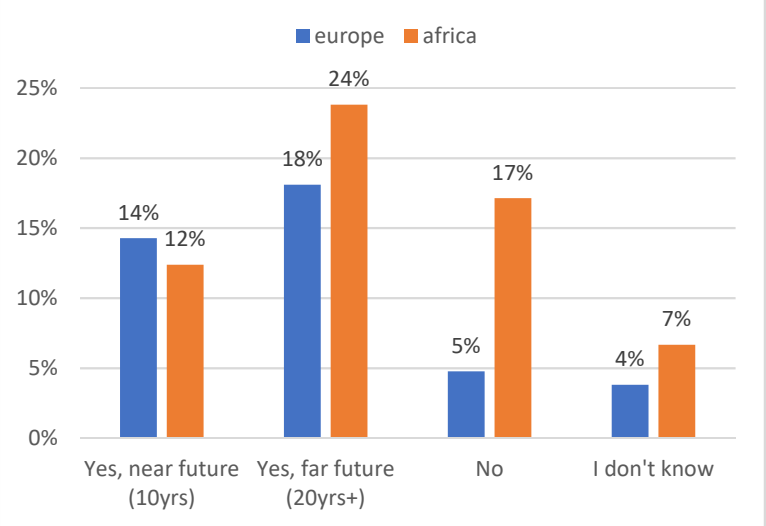
The second part of the survey was aimed to evaluate personal opinions related to the future of cash. Figure 14 shows that 14% of respondents from Europe would not be bothered at all if cash wasn't a method of payment anymore, as expected from their use of cash displayed in figure 8. 11% of respondents from Africa would be bothered a little if cash were to disappear, which is surprising considering how often they use it for certain activities shown in figure 11. However, opinions are personal and clearly vary from "a lot" to "none at all" for both African and European respondents.

Figure 14 - "How much would it bother you if you couldn't use cash anymore?" (16-25yrs)



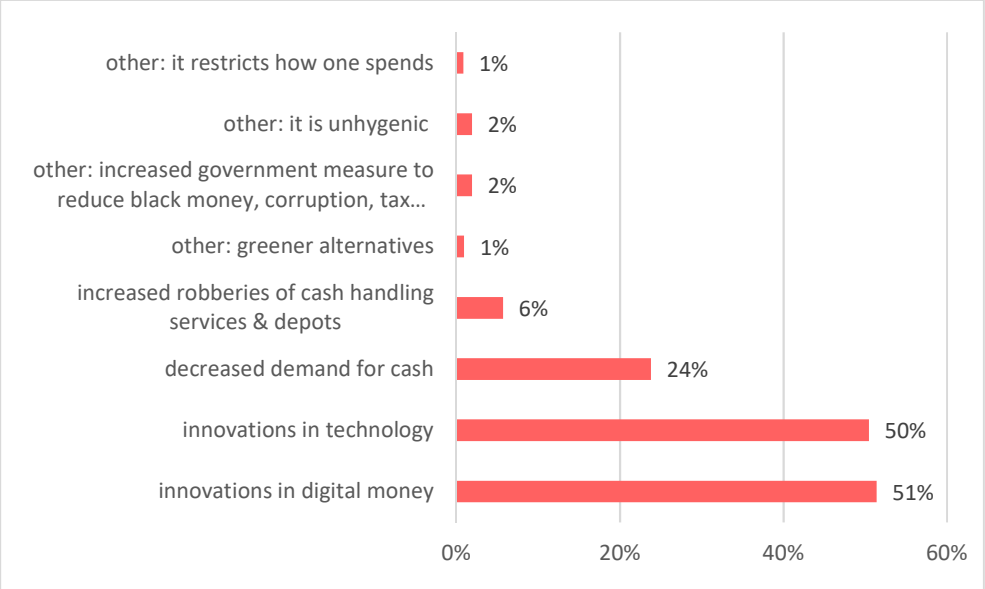
Looking at figure 15, the majority of participants believe cash will disappear in the far future, for the reasons displayed in figure 16 below.

Figure 15 - "Do you think cash will disappear?" (16-25yrs)



Since this question permitted multiple answers including a written one, the majority chose a combination of "innovations in technology" and "innovations in digital money" as primary reasons contributing to the disappearance of cash, as displayed in figure 16 below.

Figure 16 - "Yes, I think cash will disappear because..." (16-25yrs)

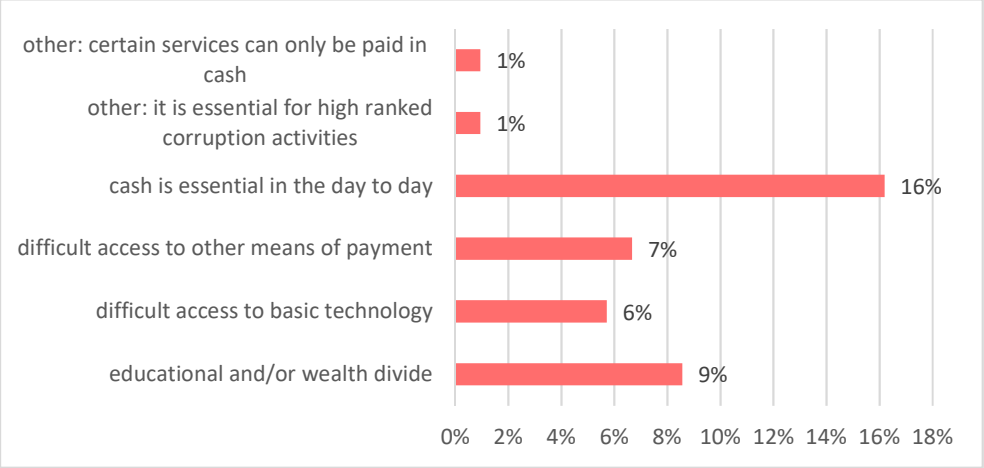


Interesting written answers included "cash is unhygienic" probably referring to the current COVID-19 pandemic which has made people generally more aware of the number of germs that can be transferred through any objects. Another written answer stipulated that there could be "greener alternatives" in the future, maybe referring to the environmental costs associated with printing, transporting and using cash.

22% of respondents who believe cash will not disappear, 17% of whom are African residents, their reasons why are displayed in figure 17.

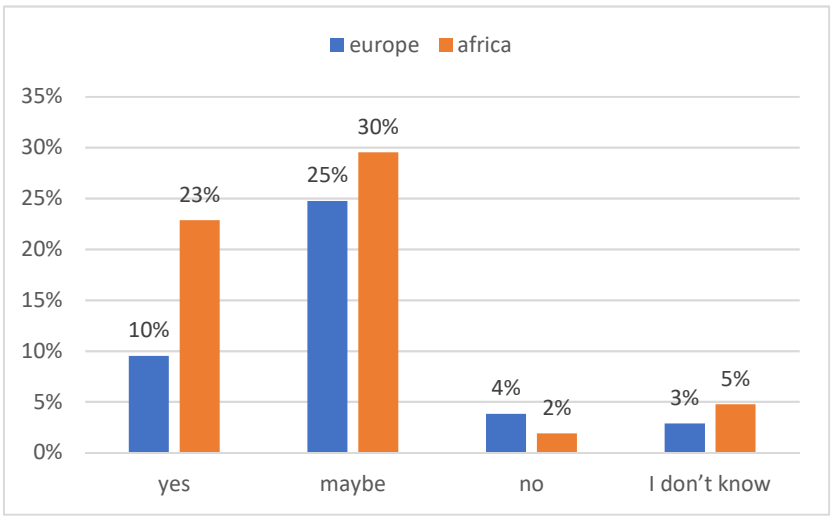
Multiple and written answers were permitted here too, where the majority of respondents chose "cash is essential in the day to day" and "educational and/or wealth divide. Written answers include the following "certain services can only be paid in cash" and "cash is essential for high ranked corruption activities" which are more relevant for nations where payment methods are underdeveloped and where corruption in both low and high positions.

Figure 17 - "No, I don't think cash will disappear because..." (16-25yrs)



Lastly, figure 18 shows what the respondents think of digital solutions possibly replacing cash in the future. 55% of respondents believe that they could replace cash; the majority of them are residing in Africa. However, 4% of European respondents disagree, as opposed to 2% of African respondents.

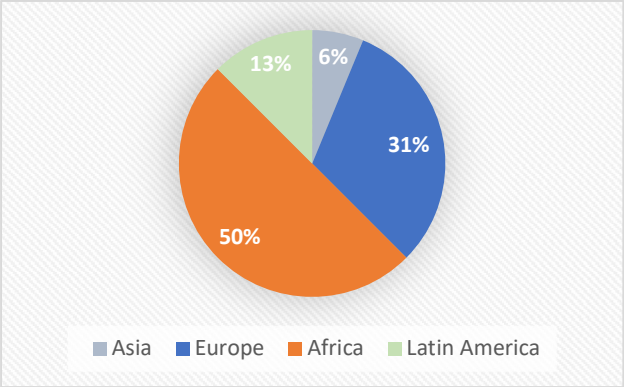
Figure 18 - "Do you think digital solutions will replace cash in the future?" (16-25yrs)



Respondents aged 26-40 (generation Y)

There were few participants from this age range, only 16, from majority Tanzania (8) and Germany (3). I decided to group the results in terms of continents for clarity, as shown in figure 19, and to only analyse responses from African and European residents. There were too few from other parts of the world; only one respondent from Asia (the Philippines) and two respondents from Latin America (Peru and El Salvador).

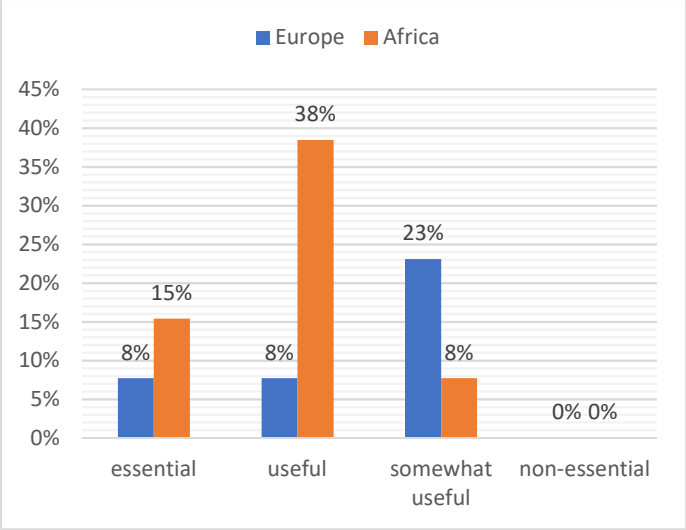
Figure 19 - Provenance of participants aged 26-40 years



As mentioned previously, there was no inquiry on the gender or occupation of the participants, only on their age and country of residence. 100% of them answered positively to having a bank account and know what online payment systems are.

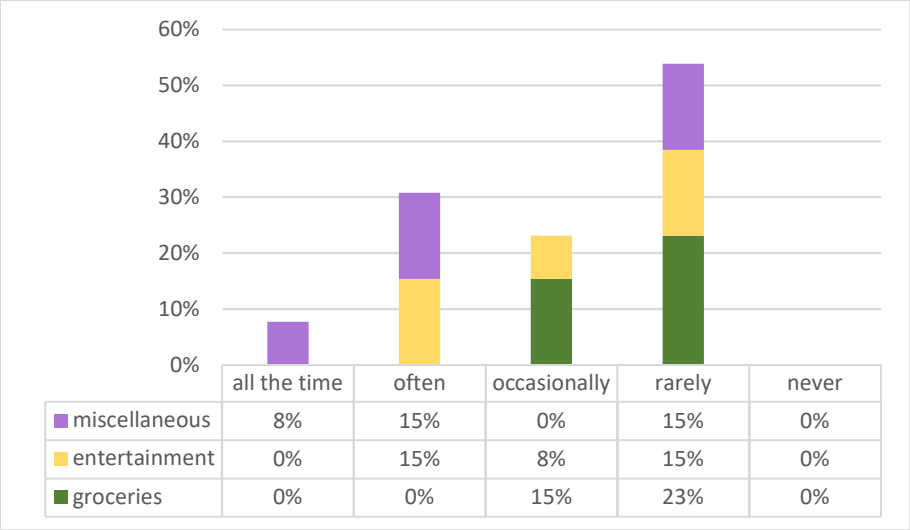
Figure 20 gives us insight into the participant's personal opinion on the usefulness of cash. Majority of answers from Africa residents deem

Figure 20 - Perceived usefulness of cash (26-40yrs)



cash essential and useful, compared to European residents categorising it as somewhat useful. This is effectively translated into the results of their payment preferences below. Figure 21 shows how often respondent from Europe use cash for different circumstances; almost all agree that they use cash rarely to purchase groceries, and often for entertainment[36] or miscellaneous expenses[37].

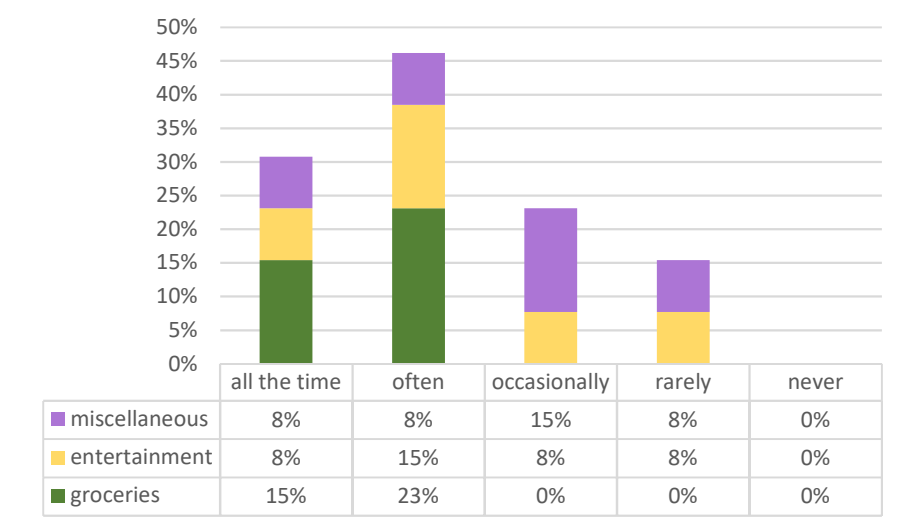
Figure 21 - Cash use Europe (26-40yrs)



	all the time	often	occasionally	rarely	never
miscellaneous	8%	15%	0%	15%	0%
entertainment	0%	15%	8%	15%	0%
groceries	0%	0%	15%	23%	0%

In figure 22, we can clearly see that respondents are more inclined to use plastic money more often or all the time for groceries and entertainment, as opposed to occasionally for miscellaneous

Figure 22 - Card use Europe (26-40yrs)



	all the time	often	occasionally	rarely	never
miscellaneous	8%	8%	15%	8%	0%
entertainment	8%	15%	8%	8%	0%
groceries	15%	23%	0%	0%	0%

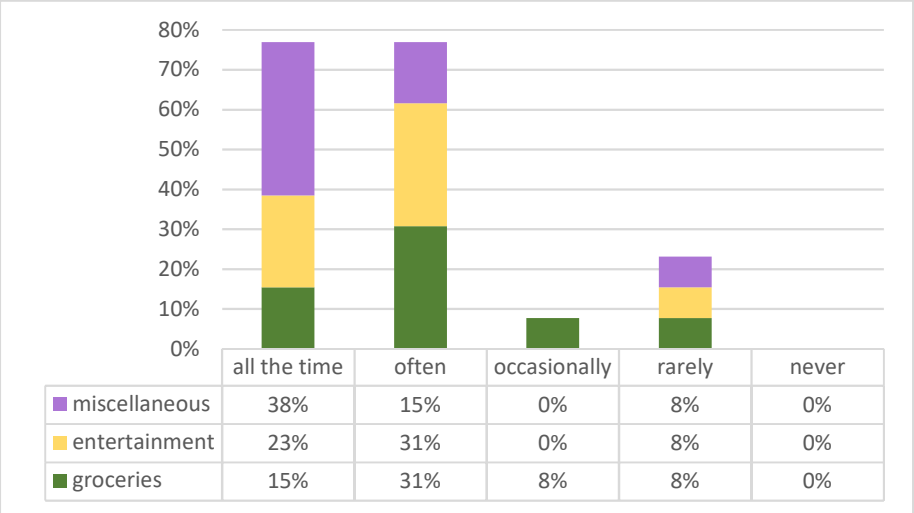
[36] cinema, bar, club, restaurant etc...

[37] Miscellaneous expenses can include (but aren't limited to), any purchases that aren't on your grocery list and which don't figure in the entertainment list mentioned above. Examples include goods like tobacco or services like cleaning.

expenses. We can assume that environmental factors are at play here; for example, it can be common to find that shops selling tobacco either don't have any card readers or require a minimum. This incentivises consumers to either spend enough in the shop to be able to pay with card, or carry cash for such expenses, and some prefer the latter.

The next two figures analyse the same questions but for African respondents. As expected, we can see that majority use cash all the time or often for all three categories, miscellaneous expenses having the most respondents using cash all the time.

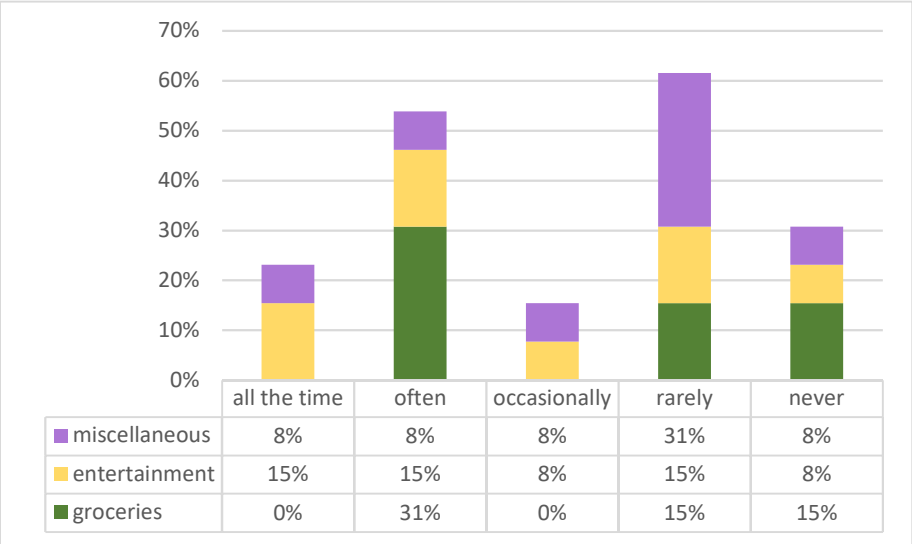
Figure 23 - Cash use Africa (26-40yrs)



	all the time	often	occasionally	rarely	never
miscellaneous	38%	15%	0%	8%	0%
entertainment	23%	31%	0%	8%	0%
groceries	15%	31%	8%	8%	0%

Figure 24 is interesting because 31% of the respondents often use card to purchase groceries, which is the same number as those who often use cash for the same category in figure 23. Card use for miscellaneous expenses is also very low, 31% say to use it rarely; card use for entertainment is inconclusive as equal parts, 15%, say to use it all the time, often and rarely.

Figure 24 - Card use Africa (26-40yrs)



	all the time	often	occasionally	rarely	never
miscellaneous	8%	8%	8%	31%	8%
entertainment	15%	15%	8%	15%	8%
groceries	0%	31%	0%	15%	15%

The next two figures review the participants' basic knowledge of digital currencies. Out of the 13 participants, 38% of whom are European all know what digital currencies are but do not own or trade any. Majority of African residents know what digital currencies are and, unexpectedly, 15% own or trade them.

Figure 26 - Know what digital currencies are (26-40yrs)

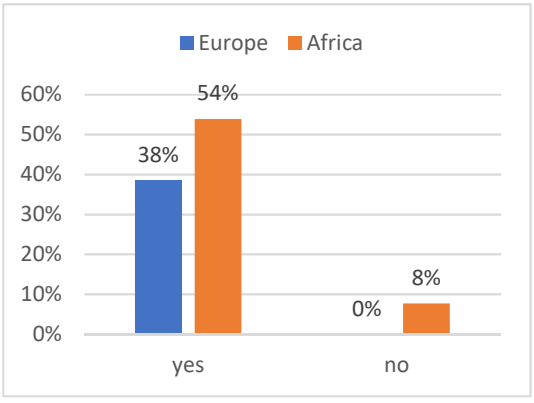
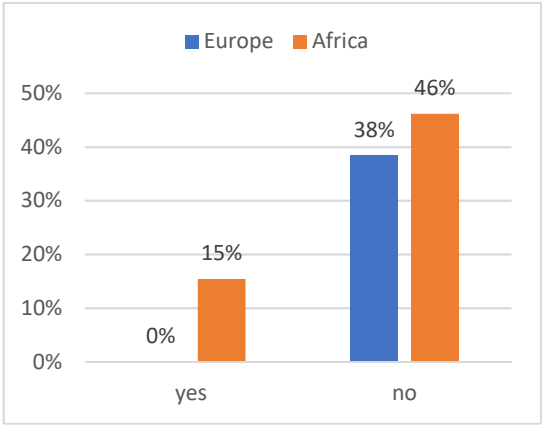
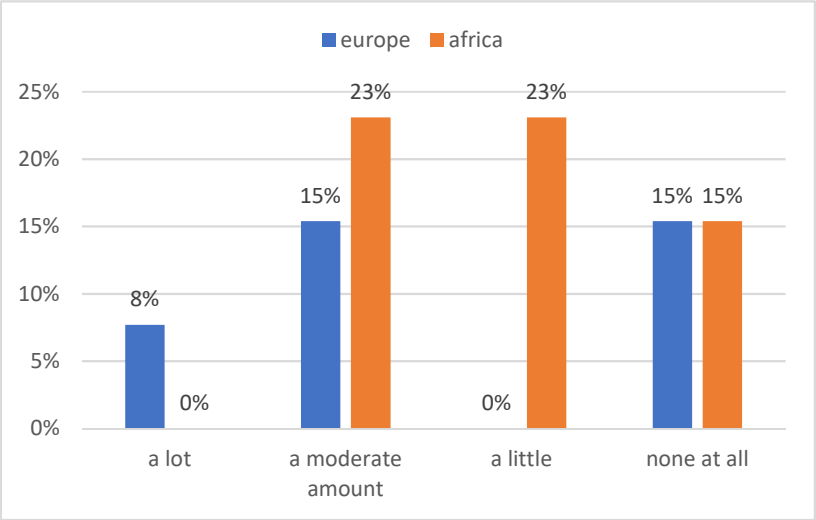


Figure 25 - Own or trade digital currencies (26-40yrs)



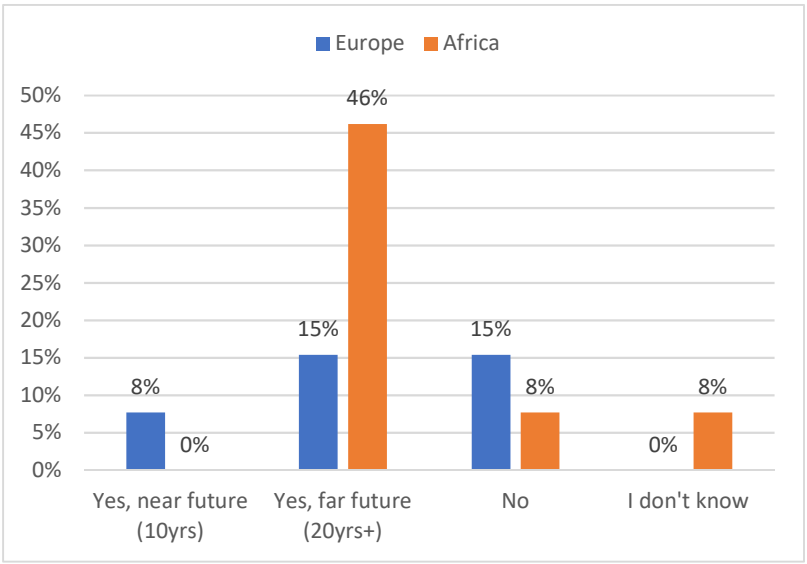
The second part of the survey was aimed to evaluate personal opinions related to the future of cash. Figure 27 shows that European respondents would mainly be moderately and none at all bothered if cash were to disappear.

Figure 27 - "How much would it bother you if you couldn't use cash anymore?" (26-40yrs)



African respondents would mainly be moderately, and a little bothered if cash wouldn't figure as a payment method anymore, which is unexpected. Still, regarding African respondents, figure 28 shows that a significant majority seem to think cash will disappear from their respective countries in the far future. On the contrary, European respondents are tied between cash disappearing in the far future and not disappearing at all.

Figure 28 - "Do you think cash will disappear?" (26-40yrs)



The following figures show why the respondents answered yes or no to cash disappearing, regardless of country of residence—this question allowed for multiple answers, including written ones.

Those who had picked either "yes" previously, in majority chose the combination "innovations in digital money" and "innovations in technology, as shown in figure 29. An interesting written answer made by one participant was "strengthened methods of surveillance", which could possibly refer to policies increasing the surveillance of cash, therefore deterring its use.

Figure 29 - "Yes, I think cash will disappear because of..." (26-40yrs)

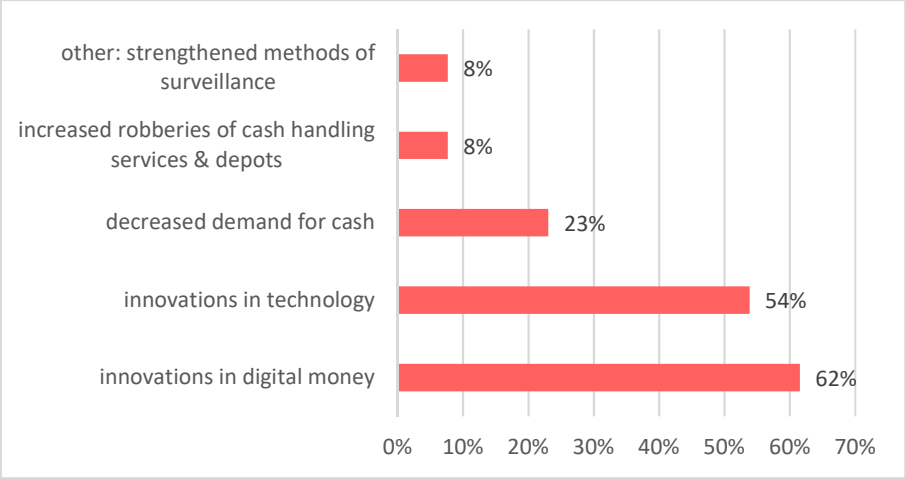
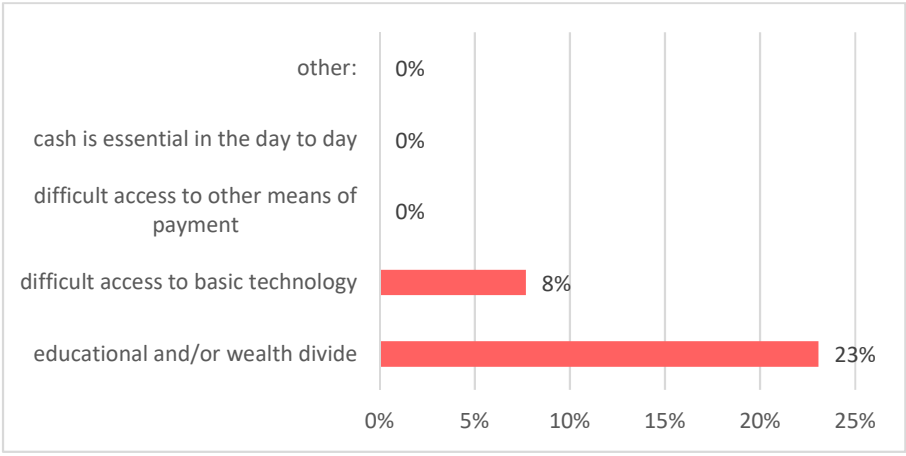


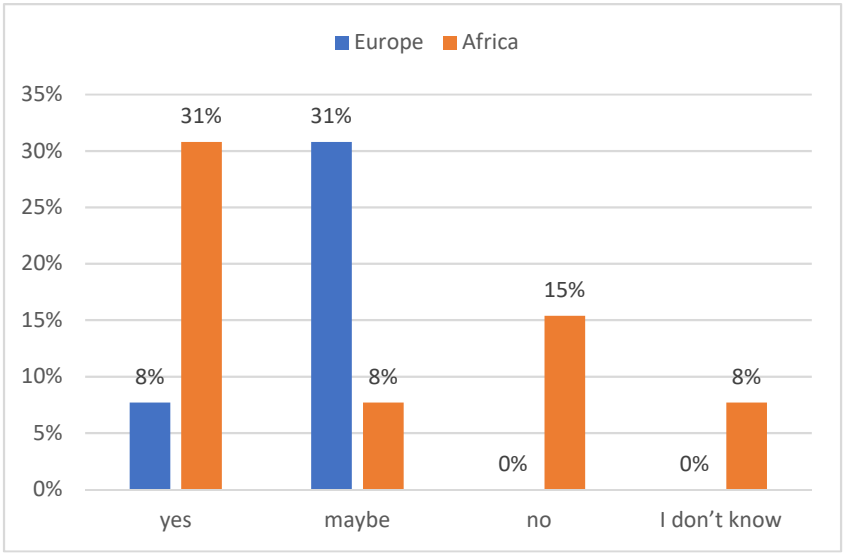
Figure 30 shows that those who had answered "no" previously both chose "educational and/or wealth divide" and "difficult access to basic technology" as a reason why cash will not disappear. Surprisingly, nobody chose "difficult access to other means of payment, which personally seemed like the most logical pick.

Figure 30 - "No, I don't think cash will disappear because of..." (26-40yrs)



Lastly, figure 31 shows how many of the respondents think digital solutions could replace cash, regardless if they picked "no". The majority of European respondents believe that they potentially could, whereas African respondents are spread out between "yes" and "no".

Figure 31 - Digital solutions to replace cash in the future (26-40yrs)



Discussion

So, what is the cashless revolution, and why is it important?

The cashless revolution refers to the rise in disruptive technologies in payment methods which don't utilise cash to settle transactions and the transition to a digitalised monetary system.

Cashless methods of payment include cards[38] which first became widely used in the 1980s, followed by the rise of mobile payments in the 2000s with M-Pesa in East Africa and Paypal in the USA, and at last, digital currencies such as Bitcoin.

Digitalisation is the process of converting something to a digital form[39], whether it be paper-based administrative records or business models. There are many benefits attributed to going digital.

In the government sector, it can simplify a country's intelligence systems and improve the delivery of public services for citizens, with mobile technologies for example. It can also help reduce embezzlement and money laundering.

For businesses, it can facilitate trade and the efficiency of their cashflows, allowing them to use their capital and their funds more effectively. More services get chained together which can help solve supply-chain inefficiencies, thus lowering the costs to operate in specific sectors. This can incentivise workers to join less popular industries like agriculture and contribute to economic growth

The digitalisation of the banking sector can provide new services to customers, such as the deposit account in Sweden, and it can aid to reduce the amount of unbanked.

Digital transformation encompasses digitalisation, and is clearly "challenging almost every aspect of the economy and society; a wide range of policy areas need to be considered as much as all the actors

[38] Debit, credit, charge etc…

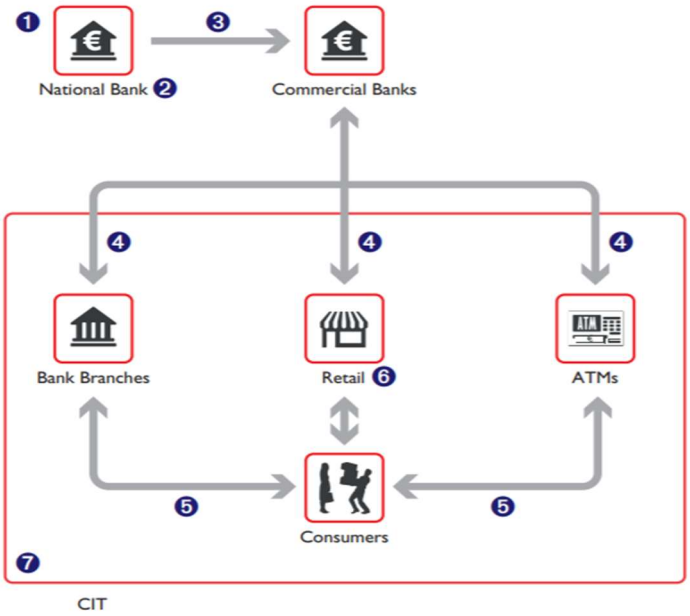
[39] Merriam-Webster 2020

(individuals, firms, governments and other stakeholders) need to be involved in the policy-making" (Reviews & Transformation, 2018).

In the case of Sweden, they are well advanced in the diffusion of digital technologies with the widespread use of internet by individuals, a relatively low digital divide across their demographic and one of the highest scores in broadband availability, affordability and quality. In 2017, Sweden launched a new digital strategy with the overall goal to achieve national digital literacy, security, innovation, leadership and infrastructure.

However, cash still has a very prominent place in our society. Based on the absolute value of ATM withdrawals and CIC around the world, demand for cash seems to be increasing steadily (Boesenach, 2018).

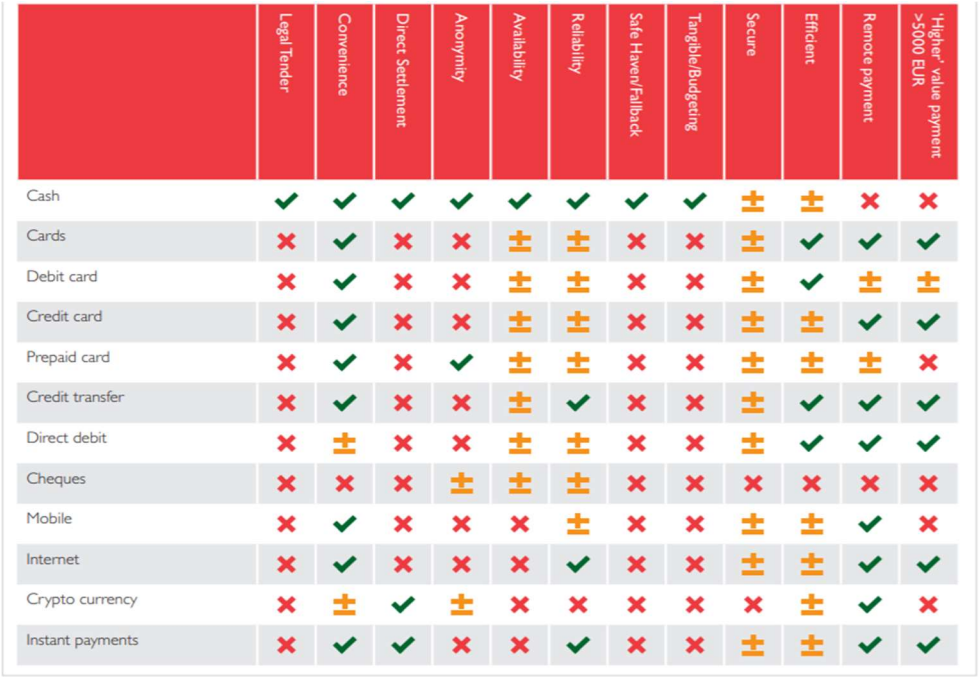
Figure 32 - Cash cycle



Also, industries very reliant on it are continuously looking for ways to optimise the Cash cycle. Figure 32 displays how the process works: national central banks determine how much cash need to be in circulation to meet public demand and replace unfit banknotes. They then produce the required amount and hand it over to cash handling companies for them to supply to commercial banks and different industries which require cash depots. Cash's unique attributes as

being reliable, available and anonymous have appealed for centuries and made it the legal tender in all countries. Figure 33 published in the World Cash Report 2018 offers a comparison between cash and competing payment methods on the attributes they do & do not possess.

Figure 33 - Scoring of payment instruments on key attributes of payments

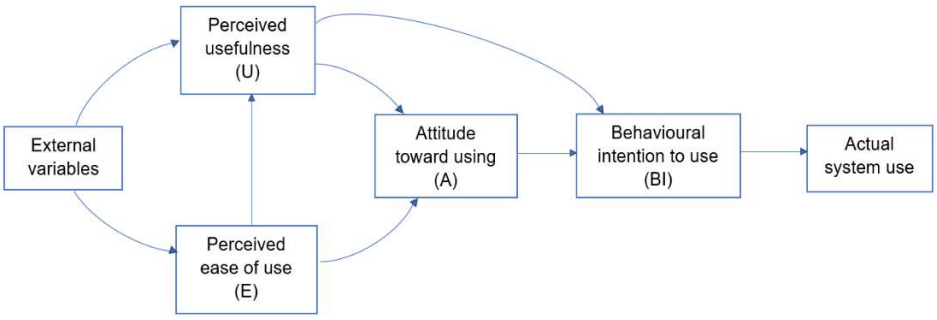


	Legal Tender	Convenience	Direct Settlement	Anonymity	Availability	Reliability	Safe Haven/Fallback	Tangible/Budgeting	Secure	Efficient	Remote payment	'Higher' value payment >5000 EUR
Cash	✓	✓	✓	✓	✓	✓	✓	✓	±	±	✗	✗
Cards	✗	✓	✗	✗	±	±	✗	✗	±	✓	✓	✓
Debit card	✗	✓	✗	✗	±	±	✗	✗	±	✓	±	±
Credit card	✗	✓	✗	✗	±	±	✗	✗	±	±	✓	✓
Prepaid card	✗	✓	✗	✓	±	±	✗	✗	±	±	±	✗
Credit transfer	✗	✓	✗	✗	±	✓	✗	✗	±	✓	✓	✓
Direct debit	✗	±	✗	✗	±	±	✗	✗	±	✓	✓	✓
Cheques	✗	✗	✗	±	±	±	✗	✗	✗	✗	✗	✗
Mobile	✗	✓	✗	✗	✗	±	✗	✗	±	±	✓	✗
Internet	✗	✓	✗	✗	✗	✓	✗	✗	±	±	✓	✓
Crypto currency	✗	±	✓	±	✗	✗	✗	✗	✗	±	✓	✗
Instant payments	✗	✓	✓	✗	✗	✓	✗	✗	±	±	✓	✓

Even so, electronic payment methods are becoming widely available, and their transaction volume is steadily growing. The analysis done in this paper complimented with the survey conducted suggests that the cashless payment revolution is well on its way; however, with some trouble in developing countries where cash is still king.

One way to understand the slow spread of cashless solutions in certain African countries such as Zimbabwe is to look at a Technology Acceptance Model (TAM). It explains how information technology is gradually adopted and accepted by people.

Figure 34 – Technology Acceptance Model



The model is determined by two pivotal beliefs: Perceived usefulness (PU) and Perceived Ease of Use (PEU). Attitudes predict intentions and intentions predict behaviour. The main argument of the model is that the adoption of a particular technology is an outcome of intention. It is determined through the perception of usefulness, ease of use, attitude to use, intention to use and actual use of said technology. The people's perceptions play a crucial role in their attitudes and behavioural intentions to use cashless alternatives (Nyoni et al., 2017). Looking at Zimbabwe, one of the reasons for the low use of cashless alternatives such as bank cards is the people's perception of a corrupt and inefficient system currently in place. That paired with low perceived ease of use and certain social limiting factors translates in the low use of bank cards.

Another reason adding to the slow transition can be explained by the theory of payment system efficiency. It states that a national central bank is poised to lose its monopoly position in the provision of liquidity, as the economy moves towards a cashless system. As demonstrated with Zimbabwe's liquidity problem and a monetary system prevalent with corruption, national central banks may not be in favour of losing their monopoly position because it may not serve their interests.
The survey conducted further suggests that cash is still very prevalent in Africa due to the widespread unavailability of cashless alternatives and an educational and wealth divide. International institutions are aware of these factors limiting populations in emerging economies to lift themselves out of poverty.

The BTCA[40] is a global partnership based under the United Nations to accelerate the transition from cash to digital solutions in a bid to reduce poverty and sustainable growth in emerging economies. The organisation, funded by Mastercard and 'The Bill and Melinda Gates Foundation' to name a few, has conducted research and worked with governments around the world to promote the use of digitalising payments in order to improve efficiency and transparency.

One of their success stories includes the '*Bolsa Família* Cash Transfer Program' enacted in 2003 in Brazil. The program combined several pre-existing social benefits programs already in place into one payment card, reducing its transaction costs by 12.1%[41], and effectively allowing to cover more than 26% of the Brazilian population (Provost, 2011). A report "Accelerators to an inclusive digital payment ecosystem" published in 2016 recommend "measures to encourage or require government entities, private businesses and individuals to shift away from cash, sometimes in policies that disincentivise cash usage" (Janis & Shah, 2016).

However, certain critics say that the BTCA encourages coercive policies, putting them in the same boat as other companies that support financial exclusion of poor people by preventing them from using their preferred and often only means of payment (White, 2018).

Coercive policies mentioned on the segment of the War on Cash were explained with the example of India's demonetisation in 2016. What these policies do is harm the livelihood of both banked and unbanked MSME's[42] who rely on cashflows to operate, they compromise the financial privacy of individuals and impose an unlegislated tax on bank account holders when a central bank pursues negative interest rates.

Although the allegations made by those critics remain unfounded, to genuinely promote the well-being of the poor and everyone else,

[40] Better Than Cash Alliance
[41] Better Than Cash Alliance - article
[42] Micro, Small and Medium Enterprises

BTCA should reject policies that restrict the choice of cash users because they do more harm than good in the end, India's demonetisation serving as the unfortunate example.

Although not mentioned a lot in this paper, it is important to note that the cashless revolution is a global phenomenon. This paper focused substantially on Europe and East Africa due to ease of collecting data, especially as regards to the survey; nevertheless, this cashless movement has been well on its way in Asia.

According to the latest World Payments Report, Asia currently figures among the top ten markets for non-cash transactions worldwide, not far behind the USA and Europe, after an unprecedented explosive increase in the span of five years. This is thanks to China's tech giants innovative blend of mobile payment, social and e-commerce all in one that revolutionised how people viewed fintech systems. The region is estimated to take over North America and Europe in terms of number of non-cash transactions worldwide in the next couple of years (Capgemini, 2019).

Conclusion

The cashless revolution is inevitable because it is already here. Everywhere, disruptive forms of payments have been introduced, and people are using them because they are convenient, they are effective, and they are helping them grow.

How would a cashless society look like? Financial transactions would be conducted through the transfer of digital information between the transacting parties (Nyoni, 2018). Central banks would have eliminated a substantial part of their costs related to taking care of cash, and illegal activities where cash is the dominant method of payment would have drastically reduced (presumably).

Many nations are still very cash-dependent, especially where access to formal banking is restricted or hard to reach. However, the elimination of cash could bring certain benefits; the reduction in money laundering, which fuels the illegal drug trade and the illicit gold and diamond mines. Overall, it would reduce the negative socioeconomic consequences associated with these activities and improve the lives of affected populations.

Participants in the illegal trade may find backdoors through the use of cryptocurrencies; just as Bitcoin provided in its early days and still does today, in addition to many other cryptocurrencies out there.

A big challenge for countries facing the entry of multiple disruptive technologies in Fintech is regulating. Policies need to protect users against exploitive charges from companies, privacy abuse and exclusion. For Sweden, groups which are unable to part with cash must be included in these cashless policies as they are most at risk of being left behind. Unfortunately for cash handling industries, any change an evolution does makes a couple of victims.

In the case of Zimbabwe, trust and stability are essential for the government to regain the trust of the people; secondly, investing in

infrastructure to improve the current cashless systems of banks and mobile payments. And at last, reduce inflation.

Regarding the research analysis and discussion, the outcome shows that younger populations in Europe are indeed using more frequently cashless payment methods in their everyday lives, as opposed to younger people residing in Africa which use cash more often. Unfortunately, due to the small size of the older respondents' sample, an accurate comparison between the two was not possible. In addition, the respondents were unfortunately concentrated in central Europe and east Africa, not allowing speaking about cashless trends in Asia, which are very prominent.

At last, the cashless revolution gives a lot of work to do to national central banks and the government, in terms of adapting their monetary policies, putting in place regulations and ensuring cohesion in the future to come.

Bibliography

Al-Laham, M., Al-Tarawneh, H., & Abdallat, N. (2009). Development of Electronic Money and Its Impact on the Central Bank Role and Monetary Policy. *Proceedings of the 2009 InSITE Conference*, *6*. https://doi.org/10.28945/3328

Arvidsson, N. (2019). *Building a Cashless Society*. https://doi.org/10.1007/978-3-030-10689-8

Arvidsson, N., Hedman, J., & Segendorf, B. (2018). *När slutar svenska handlare acceptera kontanter?* http://handelsradet.se/wp-content/uploads/2018/01/Rapport-2018_13.pdf

Boesenach, E. (2018). World Cash Report 2018. *G4S*, 196. https://www.g4scashreport.com/-/media/g4s/cash-report/files/2018-world-cash-report---english.ashx?la=en&hash=0F3BECD46B4820D7FA32112E99252AAB

Capgemini. (2019). *World Payments Report 2019*. 48.

Cochrach, M. (2019). *The History of the War on Cash*. 8. https://doi.org/10.1017/s0001924000150456

Details, P., & Transactions, P. (2020). *Global Cards*. 1–2.

Esselink, H., & Hernández, L. (2017). The use of cash by households in the euro area. *Occasional Paper Series*, *201*. https://doi.org/10.2866/377081

Federal Reserve Bank of Dallas. (2011). Hyperinflation in Zimbabwe. In *Globalization and Monetary Policy Institute 2011 Annual Report* (Issue Bernholz 2003, pp. 2–12).

Federal Statistical Office Germany. (2020). *Bevölkerungsstand - population status*. https://www.destatis.de/DE/Themen/Gesellschaft-Umwelt/Bevoelkerung/Bevoelkerungsstand/Tabellen/zensus-

geschlecht-staatsangehoerigkeit-2019.html

Gawra, T. (1996). *History of tokens*. 2–7.

Hogg, C. (Visa C. E. (2018). *Visa Response to June 2018 outage*.

Janis, W., & Shah, R. (2016). *Accelerators to an Inclusive Digital Payments Ecosystem* (Issue september). https://btca-prod.s3.amazonaws.com/documents/238/english_attachments/BTC_Accelerator_Report.pdf?1474430605

Lewis, P., & Kozhani, R. (2020). *Visa' s European payment systems back up after outage*. 2–3.

Manur, A. (2016). *We are trying to become a cashless society — but is that a great idea ?* 1–7. https://cashessentials.org/news/we-are-trying-to-become-a-cashless-society-but-is-that-a-great-idea/

Mbiti, I., Weil, D. N., Edwards, S., & Johnson, S. (2016). The Impact of M-Pesa in Kenya. In *African Successes, Volume III: Modernization and Development* (pp. 247–293). University of Chicago Press.

Zimbabwe: A country study, (1983).

Nyoni, T. (2018). Towards a Cashless Zimbabwe: an Empirical Analysis. *International Journal of Marketing & Financial Management*, *6*(1), 32–44. file:///C:/Users/msimatele/Downloads/ARSEAM-1.pdf

Nyoni, T., Bonga, W. G., & Sakarombe, U. (2017). *Electronic Money or Cash ? In Face of Liquidity Crisis in*. *2*(11), 63–73. https://doi.org/10.6007/IJARBSS/v7-i11/3433

Perekwa, G., Prinsloo, T., & Van Deventer, J. (2016). *The impact of mobile technology on micro and small enterprises in Zimbabwe in the post-hyperinflation economic era*. *8*(3).

Provost, C. (2011). *Social security is necessary and globally affordable , says*. 1–3.

Reviews, O., & Transformation, D. (2018). *Going Digital in Sweden OECD Reviews of Digital Transformation*.

Sverigerisksbank. (2019). *Payments in Sweden 2019*. https://www.riksbank.se/globalassets/media/rapporter/sa-betalar-svenskarna/2019/engelska/payments-in-sweden-2019.pdf

Sweden Statistics. (2020). *Population density per sq. km by sex, observations and year* (p. 1).

Tett, G. (2020). *Debt : it' s back to the future*. 9–11.

White, L. H. (2018). The Curse of the war on cash. *Cato Journal*, *38*(2), 477–487. https://www.cato.org/cato-journal/springsummer-2018/curse-war-cash

World Bank. (2012). *FinScope MSME Survey Zimbabwe 2012 94505*. http://documents.worldbank.org/curated/en/780081468137402417/Zimbabwe-FinScope-MSME-survey-2012

Definitions

Merriam Webster Dictionary 2020

Money – page 5 of this paper https://www.merriam-webster.com/dictionary/money

Barter – page 5 of this paper https://www.merriam-webster.com/dictionary/barter

Mint – page 5 of this paper https://www.merriam-webster.com/dictionary/mint

Cryptocurrencies – page 8 of this paper https://www.merriam-webster.com/dictionary/cryptocurrencies

Digitalisation – page 52 of this paper https://www.merriam-webster.com/dictionary/digitalization

Investopedia

History of Money – page 5 & 7 of this paper

https://www.investopedia.com/articles/07/roots_of_money.asp

Fiat Money – page 16 of this paper

https://www.investopedia.com/terms/f/fiatmoney.asp

Hyperinflation – page 19 of this paper

https://www.investopedia.com/terms/h/hyperinflation.asp

Monetary policy – page 19 of this paper

https://www.investopedia.com/terms/m/monetarypolicy.asp

Liquidity trap – page 20 of this paper

https://www.investopedia.com/terms/l/liquiditytrap.asp

Lexico

Black Money – page 23 of this paper

https://www.lexico.com/definition/black_money

The Balance

Automated Clearing House – page 12 of this paper

https://www.thebalance.com/what-does-ach-stand-for-315226

Additional sources

Wikipedia

Numbers for Vodacom Users mentioned page 13 of this paper; original source was unavailable https://en.wikipedia.org/wiki/Vodacom_Tanzania

Lancaster House Agreement mentioned page 27 of this paper; original source unavailable
https://en.wikipedia.org/wiki/Land_reform_in_Zimbabwe

World Bank Data

Numbers about Zimbabwe's employment in agriculture and agriculture % of GDP mentioned page 27 of this paper; graphs in appendix
https://data.worldbank.org/country/zimbabwe

Reserve Bank of Zimbabwe

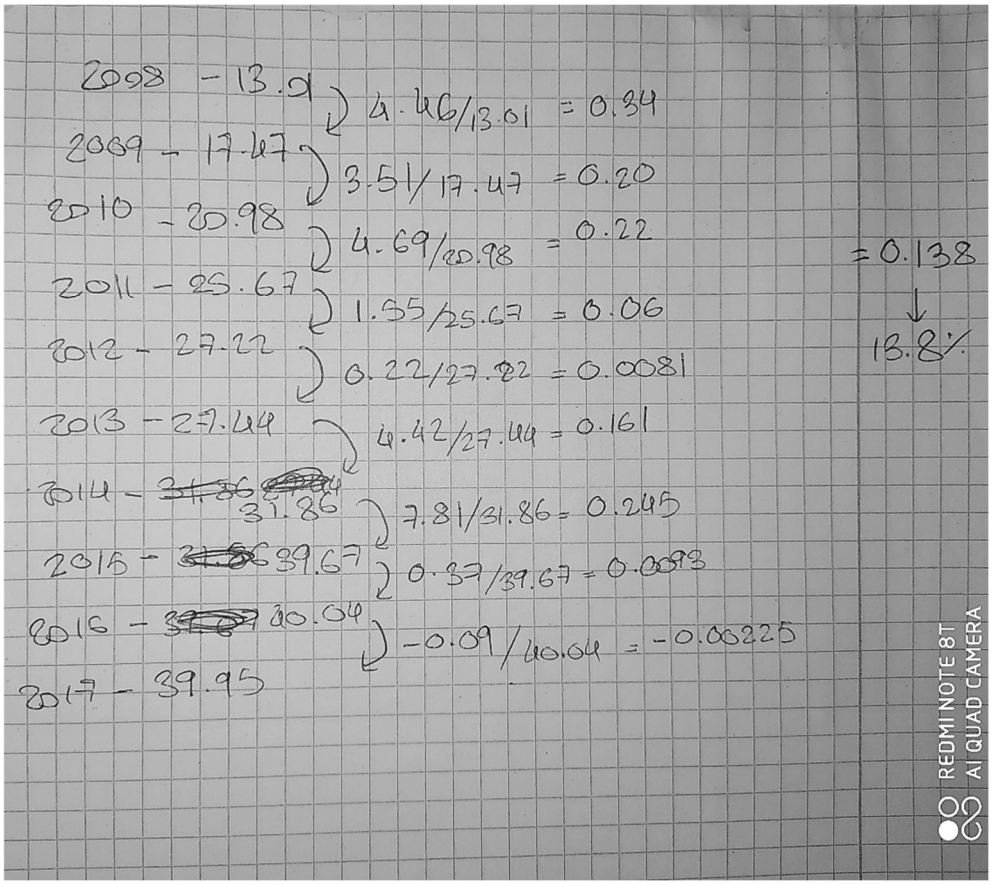
Official Data about inflation for March 2020, mentioned page 33 of this paper (source on behalf of Trading Economics)
https://tradingeconomics.com/zimbabwe/inflation-cpi

Better Than Cash Alliance

Explanation of one of their projects mentioned page 50 of this paper
https://www.betterthancash.org/why-digital-payments/governments

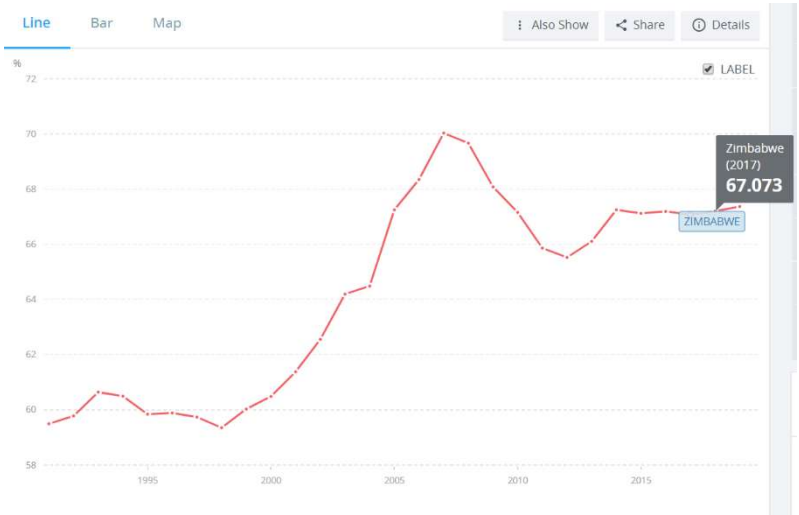
Appendix

1. Calculations mentioned in page 13 based on data from "The Impact of M-Pesa in Kenya" by Mbiti 2016

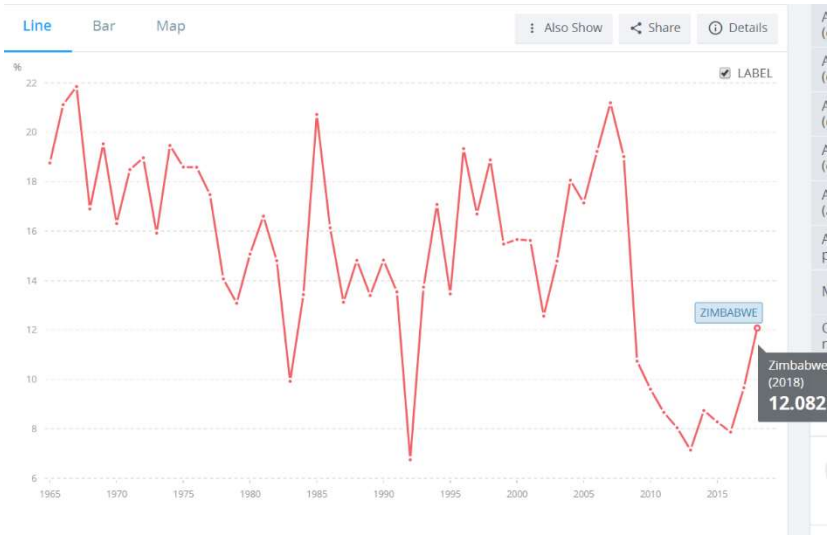


2. World Bank Data 2018 - Graphs relevant for pg 27 of this paper

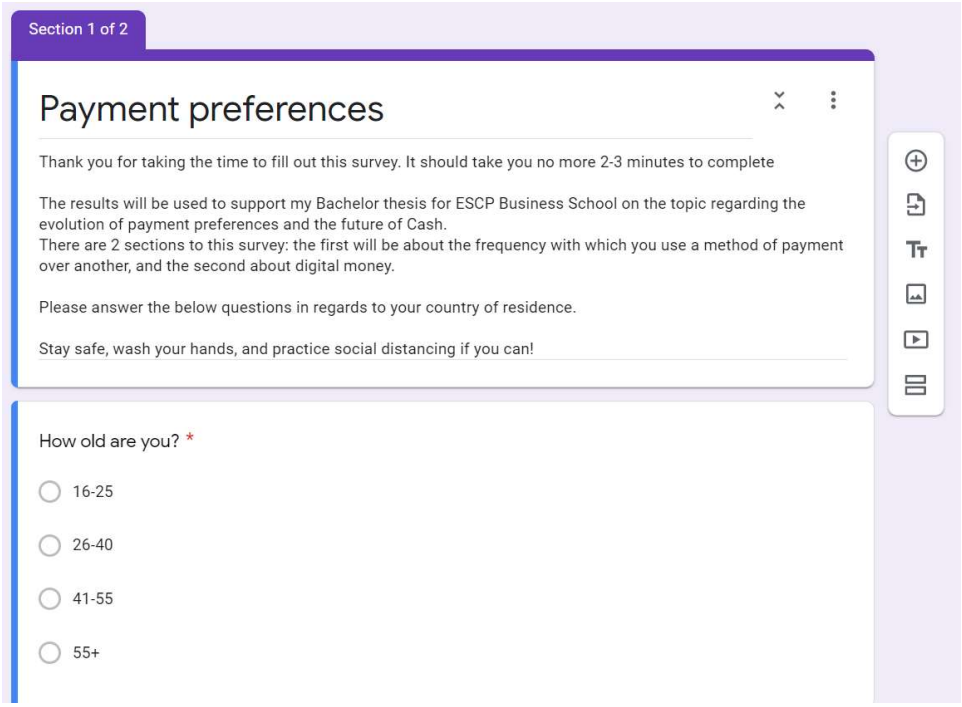
Population % employed in the agricultural sector (Zimbabwe)



Agriculture as % of GDP (Zimbabwe)



3. Survey conducted





Do you know what online payment systems are? *

- Yes
- No
- Unsure

Title

An online or e-commerce payment system facilitates the acceptance of electronic payment for online transactions.

A mobile payment system allows for payment for goods or services through a portable electronic device such as a tablet or a cell phone.

Have you ever used one of the following systems? (If no, tick 'Other' and type NA) *

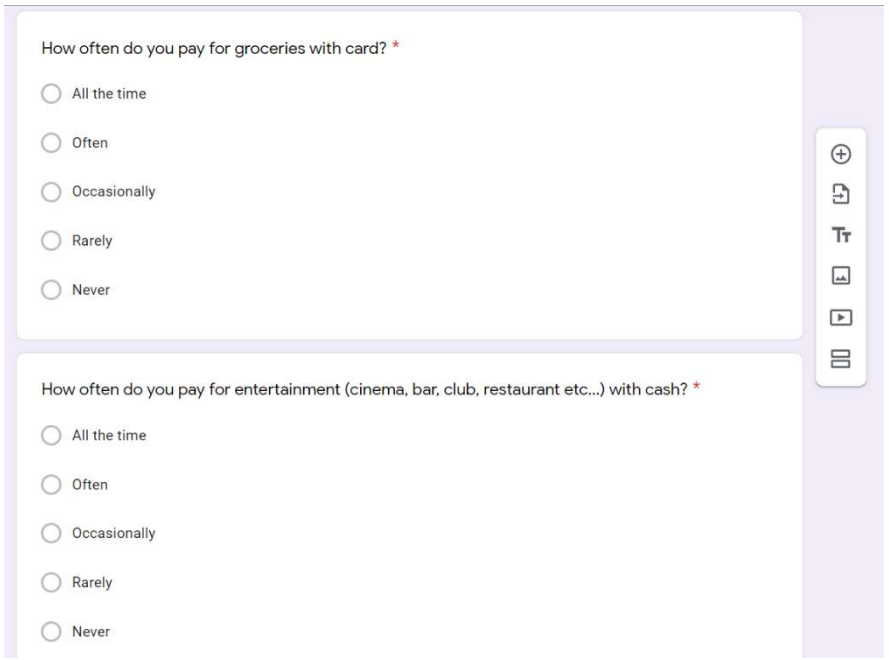
- [ ] Apple Pay
- [ ] Google Pay
- [ ] Klarna
- [ ] M-pesa
- [ ] Paypal
- [ ] WePay
- [ ] Other...

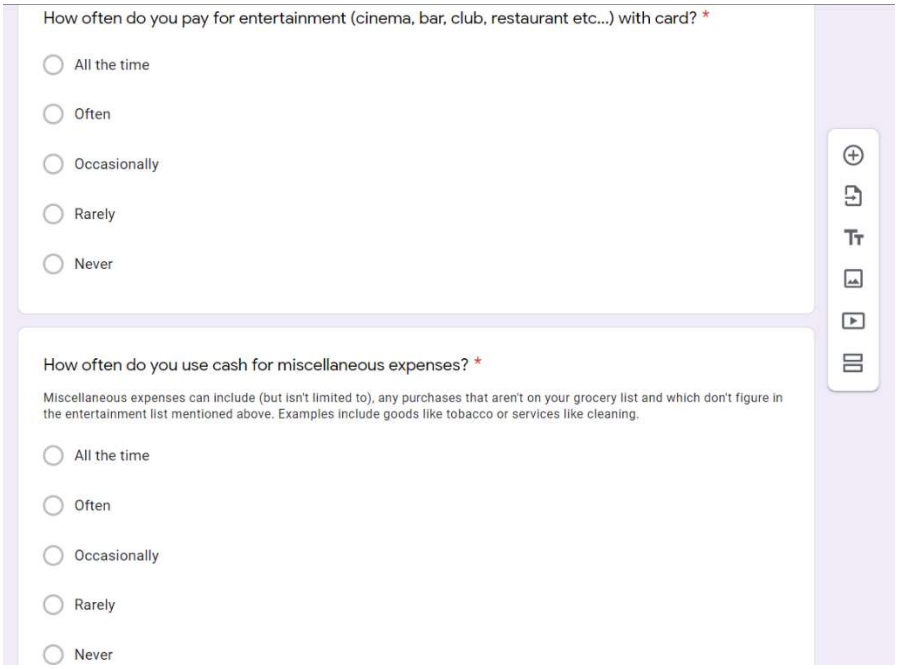
To what extent do you consider cash useful in today's society? (In regards to your country of residence) *

- Essential
- Useful
- Somewhat useful
- Non-essential

How often do you pay for groceries with cash? *

- All the time
- Often
- Occasionally
- Rarely
- Never


How often do you pay for groceries with card? *
All the time
Often
Occasionally
Rarely
Never
How often do you pay for entertainment (cinema, bar, club, restaurant etc...) with cash? *
All the time
Often
Occasionally
Rarely
Never


How often do you pay for entertainment (cinema, bar, club, restaurant etc...) with card? *
All the time
Often
Occasionally
Rarely
Never
How often do you use cash for miscellaneous expenses? *
Miscellaneous expenses can include (but isn't limited to), any purchases that aren't on your grocery list and which don't figure in the entertainment list mentioned above. Examples include goods like tobacco or services like cleaning.
All the time
Often
Occasionally
Rarely
Never

How often do you use card for miscellaneous expenses? *

Miscellaneous expenses can include (but isn't limited to), any purchases that aren't on your grocery list and which don't figure in the entertainment list mentioned above. Examples include goods like tobacco or services like cleaning.

- ○ All the time
- ○ Often
- ○ Occasionally
- ○ Rarely
- ○ Never

Are you aware of some of the problems cash causes and/or contributes to? *

- ○ Yes
- ○ No
- ○ Unsure

Title

Handling, transporting, printing and using cash has caused or contributed to the following issues:
- Counterfeiting
- Robberies and theft
- Inflation
- Accounting errors
- Tax avoidance & evasion
- Market for illegal goods, services & substances

How much would it bother you if you couldn't use cash anymore? *

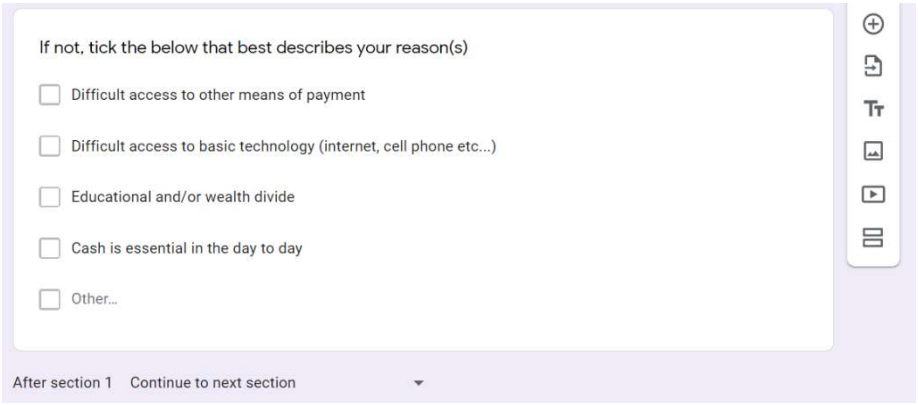
- ○ A lot
- ○ A moderate amount
- ○ A little
- ○ None at all

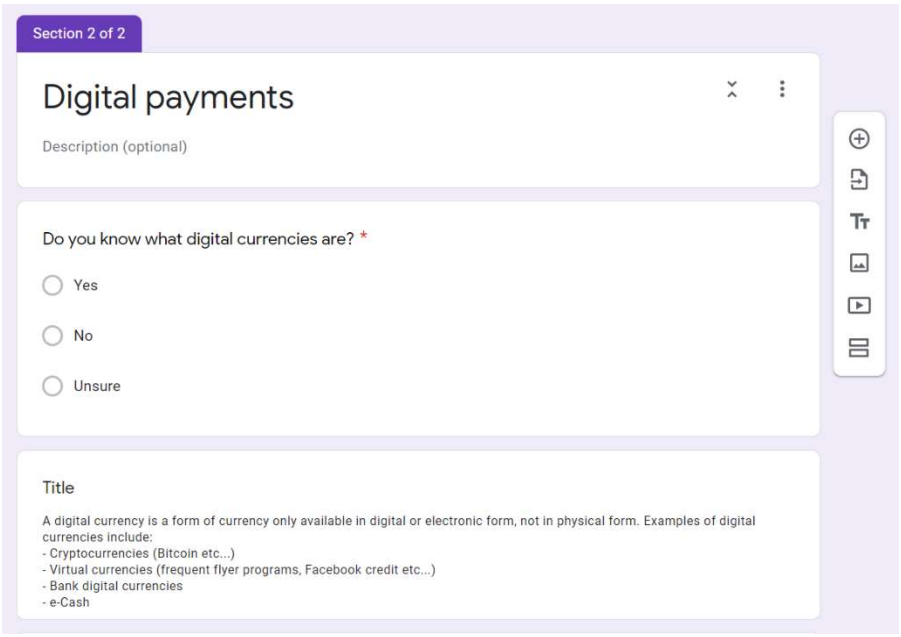
Do you think cash will disappear? *

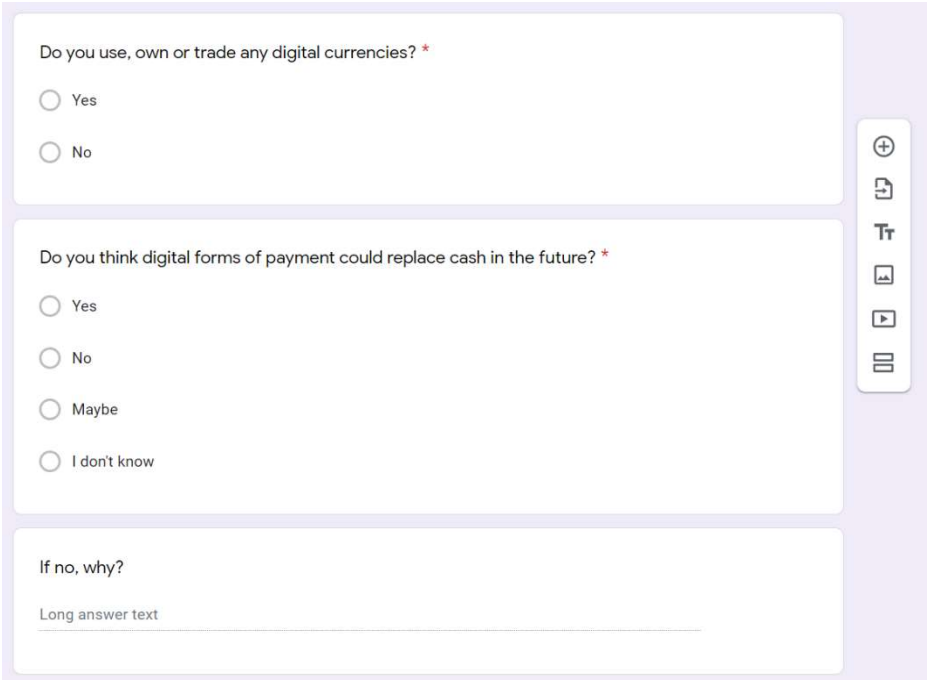
- ○ Yes, in the near future (10 years)
- ○ Yes, in the far future (20 years+)
- ○ No
- ○ I don't know

If yes, tick the below that best describes your reason(s)

- ☐ Innovations in digital money
- ☐ Innovations in technology
- ☐ Decreased demand for cash
- ☐ Increased robberies of cash handling services & depots
- ☐ Other...


If not, tick the below that best describes your reason(s)
Difficult access to other means of payment
Difficult access to basic technology (internet, cell phone etc...)
Educational and/or wealth divide
Cash is essential in the day to day
Other...
After section 1 Continue to next section


Section 2 of 2
Digital payments
Description (optional)
Do you know what digital currencies are? *
Yes
No
Unsure
Title
A digital currency is a form of currency only available in digital or electronic form, not in physical form. Examples of digital currencies include:
- Cryptocurrencies (Bitcoin etc...)
- Virtual currencies (frequent flyer programs, Facebook credit etc...)
- Bank digital currencies
- e-Cash


Do you use, own or trade any digital currencies? *
Yes
No
Do you think digital forms of payment could replace cash in the future? *
Yes
No
Maybe
I don't know
If no, why?
Long answer text

The sworn declaration

Sworn Declaration

Being aware that making a false declaration is an offence punishable by law and being aware of the legal consequences of this for the examination process, which go as far as exclusion from it in the event of cheating or the attempt to cheat in accordance with the Examination Rules of ESCP Europe Wirtschaftshochschule Berlin e.V. ("ESCP Europe"), I hereby undertake the following sworn declaration to ESCP Europe:

I have written the thesis entitled “The Global Cashless Payment Revolution: A look at Europe and Africa”

entirely by myself and independently and have used no other than the sources and resources named. Further, all the quotations and citations – including tables, maps and figures – are clearly marked as such. This declaration includes the data accessed on the internet. The thesis has not been presented to any other examining body; nor has it been published in whole or in part.

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